

TM Forum Specification

User Role Permission Management API User Guide

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Introduction

This document specifies the REST API for User Roles and Permissions Management, providing the formal definition of the underlying data model along with the complete set of operations required for the creation, assignment, and validation of user permissions. The API is designed to enable fine-grained access control by governing user roles and the authorization rules associated with them.

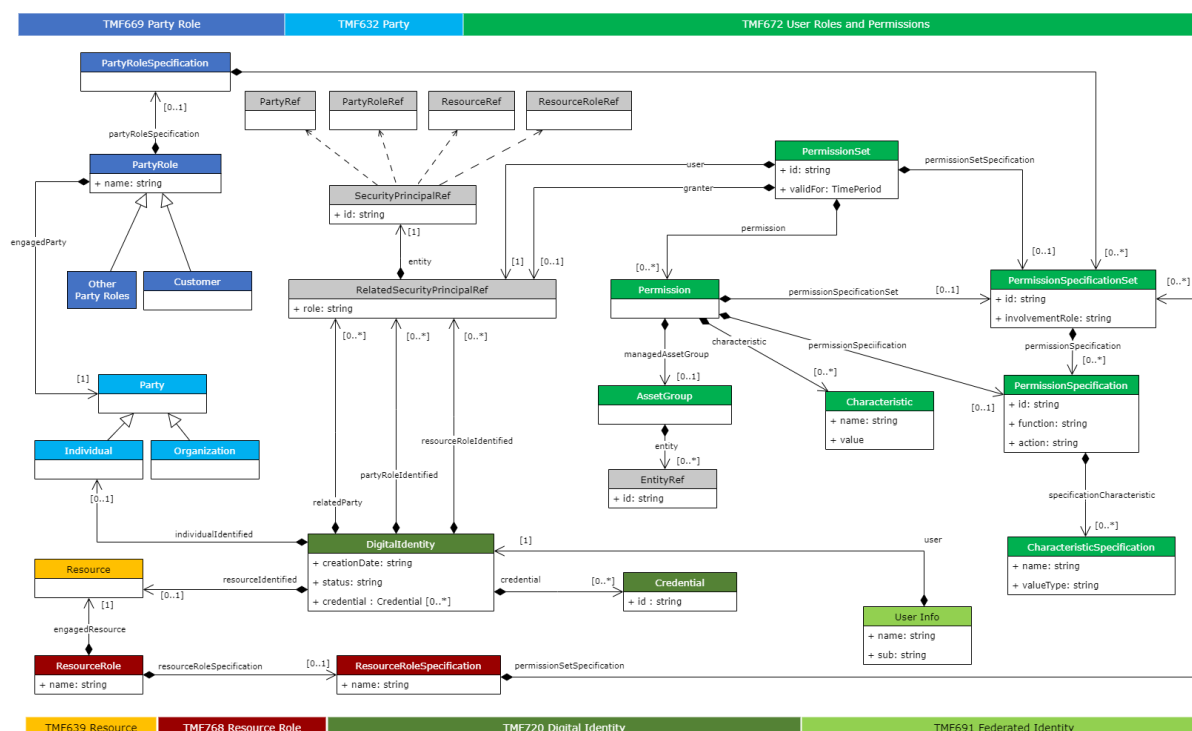
As part of the TM Forum Open API ecosystem, this API has a critical role within the broader Party, Resource and Access Management domain. It complements and integrates with related specifications, including:

- TMF632 Party Management API
- TMF669 Party Role Management API
- TMF629 Customer Management API
- TMF720 Digital Identity Management API
- TMF691 Federated Identity API
- TMF639 Resource Management API
- TMF768 Resource Role Management API

Collectively, these APIs establish a standardized model for managing parties, resources, their roles, and their associated digital identities across organizational and network boundaries. The User Roles and Permissions Management API extends this framework by providing the mechanisms required to enforce authorization and access control policies over data and resources.

Through standardization, the API ensures interoperability with other TM Forum Open APIs and external systems, supporting consistent, scalable, and secure integration patterns for identity, role, and access management across heterogeneous environments.

The diagram below illustrates the entities involved in the access control domain, showing their relationships with the corresponding TM Forum Open APIs and associated resource model entities.:



This API provides a set of operations for managing permissions, permission sets, and permission specification sets, as well as for verifying access rights.

• Permission Sets

- **Create** - Define a new permission set that grants a security principal access to manageable assets owned by an individual.
- **Update** - Revise an existing permission set, either fully or partially.
- **Delete** - Remove an existing permission set.
- **Retrieve (Single)** - Access detailed information about a specific permission set.
- **Retrieve (Multiple)** - View a list of permission sets, with optional filtering (e.g., by date created, granter, user, or asset).

• Permission Specification Sets

- **Create** - Establish a new specification set that outlines permissions, which can be linked to a party role specification or referenced in a permission set.
- **Update** - Modify an existing specification set, either fully or partially.
- **Delete** - Remove a specification set.
- **Retrieve (Single)** - Access details of a specific specification set.
- **Retrieve (Multiple)** - List available specification sets, with optional filters for easier discovery.

• Permission Specifications

- **Create** - Define a new individual permission specification.
- **Update** - Update an existing specification, either fully or partially.
- **Delete** - Remove a permission specification.

- *Retrieve (Single)* - Access details of a specific permission specification.
- *Retrieve (Multiple)* - View a list of existing specifications, with optional filters.

- **Permission Validation**

- *Check Permission* - Verify whether a given security principal has the right to perform an action defined in a permission specification, optionally tied to a specific asset or permission characteristic.

Access to this API must use a secure mechanism (e.g., OAuth 2.0), ensuring that permissions to manage assets are granted only to consumers who possess valid authorization to perform operations or delegate permissions on those assets.

This API is based on the assumption that, when a customer purchases a product under a specific account, the product instantiation carried out during the provisioning process may result in the creation of one or more Manageable Assets (e.g., registration in an eCare system, the creation of a digital service account, etc.).

Each Manageable Asset generated in this way is automatically associated with an owner, defined as the individual who holds administrative rights over the customer and account context in which the product was purchased. This initial association establishes the root permission on the asset — granting ownership access to that individual.

In cases where the purchase process designates a different administrator, this alternate individual will receive the root permission instead. From that point onward, the asset owner is empowered, through this API, to delegate access to other users by assigning additional permissions with varying access levels.

Terminology

For this API, the following definitions are provided as normative references. This version introduces some updated terminology to ensure consistency with the Information Framework (SID) data model. Where terminology differs from the previous release, the earlier terms are explicitly referenced within the definitions to maintain backward traceability and assist with migration from earlier versions.

Manageable Asset

A *Manageable Asset* is any digital or physical entity that can be provisioned, monitored, and controlled by users or systems. Examples include network resources, digital service accounts, customer data records, or temporary entities (e.g., a shopping cart).

- In IAM terms, it is a *Protected Resource* to which access control policies apply.
- Assets are tracked as managed entities in the TM Forum model, with a lifecycle subject to ownership, administrative rights, and delegated permissions.
- Under modern standards (e.g., Zero Trust), every request to access a Manageable Asset requires continuous authentication, authorization, and conditional policy evaluation.

Asset Group

An *Asset Group* is a logical collection of *Manageable Assets* organized by defined criteria.

- Groups may be based on filters (e.g., query expressions), JSON Path rules, explicit lists (IDs), or sets of predefined entity categories (e.g., all catalog items owned by a customer).
- In IAM standards (e.g., NIST RBAC/ABAC), *Asset Groups* enable policy enforcement at scale, allowing permissions to be applied consistently across multiple resources.
- This abstraction supports bulk authorization decisions, policy scoping, and delegation for both human and machine users.

Permission

A *Permission* is an explicit authorization statement allowing a *Security Principal* (user or system) to perform an action (e.g., read, write, administer) on a resource or asset.

- Permissions can be global (not tied to a specific entity) or scoped to one or more entities.
- They map directly to access control policies in frameworks like RBAC (*Role-Based Access Control*) or ABAC (*Attribute-Based Access Control*).
- In Zero Trust, each permission is continuously validated before access is granted.

Permission Set

A *Permission Set* is a collection of permissions granted to a *Security Principal*.

- *Permission Sets* may be assigned explicitly by an administrator or implicitly via the role(s) a user or resource holds.
- They may include a validity period, ensuring permissions are time-bound and revocable

when required (aligning with least privilege and NIST lifecycle accountability).

- In practice, *Permission Sets* translate roles, contexts, and attributes into enforceable access policies, enabling both role-based and dynamic attribute-based control.

Permission Specification

A *Permission Specification* defines the scope of what a permission allows:

- The *Action* (e.g., read, write, grant, revoke).
- The *Function* (e.g., entity type such as Customer, or system-level operations such as grantAlteration).
- This forms the blueprint of authorization, defining what activity is permissible on which resource.
- It aligns with policy definition in standards like XACML, where request decisions are based on action–resource combinations.

User

A *User* is a digital identity representing any entity—human or non-human—that can be authenticated and granted permissions to access resources. Users include:

- *Human Users*: Individuals such as employees, customers, contractors, or partners
- *Non-Human Users*: Applications, services, devices, IoT sensors, automated workloads, or AI agents that require system access

Security Principal

A *Security Principal* is the fundamental identity construct in IAM representing any entity subject to authentication, authorization, and audit. It may be:

- A *Human Principal* - an individual interacting with the system using credentials or federated identity.
- A *Non-Human Principal* - an autonomous process, application, API client, service account, or IoT device, authenticated via tokens, certificates, or secure keys.
- The *Security Principal* is the trust anchor in IAM and Zero Trust models, ensuring all actions can be attributed, authenticated, and authorized. It underpins identity governance across people and systems.

Role-Based Access Permission (RBAC)

RBAC is an access control model in which the activities a *Security Principal* (user or system) can perform—and the resources they can access—are determined by the roles assigned to them.

- A role represents a predefined set of permissions that correspond to specific job functions, organizational responsibilities, or system activities.
- By acquiring a role, a *Security Principal* automatically inherits all permissions associated with that role.

- This scheme enforces the least privilege principle, as principals are only granted access relevant to their assigned role(s).
- RBAC simplifies administration and improves consistency by managing permissions at the role level instead of individually per user.
- In modern IAM frameworks, RBAC can be combined with Attribute-Based Access Control (ABAC) or Policy-Based Access Control (PBAC) to provide more dynamic and context-aware decision making (e.g., validating time, location, or device type in addition to role).

Attribute-Based Access Permission (ABAC)

ABAC is an access control model that extends RBAC by incorporating attributes (characteristics or properties) of the *Security Principal*, the resource (asset), and the context of the request into access decisions.

- Attributes may include properties such as user role, department, security clearance level, device trust level, resource ownership, data classification, time of day, or geolocation.
- Access control policies evaluate these attributes dynamically to determine whether a specific action (e.g., read, update, delete) is permitted.
- ABAC supports fine-grained, context-aware authorization, overcoming RBAC's limitations by considering dynamic conditions beyond static roles.
- It is recognized in IAM standards (e.g., NIST SP 800-162) as a key enabler for Zero Trust architectures, where continuous policy evaluation is required.

Asset Group

In some scenarios, it becomes necessary to define a collection of entities. This definition can be made either explicitly (by directly listing which entities are included) or implicitly (by specifying rules or conditions that determine which entities fall into the collection). A few examples include:

- *Customer-owned entities*: Suppose a customer owns a set of entities (such as accounts, records, or assets). That customer has full rights over their entities, but those entities are not accessible to other customers. Here, the grouping of entities is based on ownership.
- *Business-specific product catalog*: A product catalog might contain many entities, but only those belonging to a specific business line (say, the mobile division). A marketer responsible for that line can add or update catalog items, but not for other business areas. In this case, the grouping is defined by business category.
- *Delegated product access*: A customer might allow someone else (a delegate, such as an assistant or partner) to access only a particular list of products. The delegate's permissions are confined to those exact items, not the customer's entire portfolio. This is an explicit, itemized list of entities.

Checking Permissions

In addition to enabling the creation and assignment of permissions, this API offers a dedicated operation for real-time permission validation. This allows clients to directly query whether a specific action is authorized at the moment it is requested, providing a more streamlined and secure alternative to retrieving and processing the entire set of permissions for a security principal on the client side.

For straightforward Role-Based Access Control (RBAC) scenarios, an API gateway can invoke this runtime check and cache the outcome for performance efficiency. For more complex access control requirements—such as context-aware or dynamic permissions—the operation can instead be called from an experience or business logic layer, enabling access decisions that reflect both business policy and runtime context.

The following example demonstrates how to use this API to determine, in real time, if a product manager is entitled to create new product offerings, illustrating its applicability for fine-grained, on-demand authorization.

```
POST /rolesAndPermissionsManagement/v5/checkPermission HTTP/1.1
Content-Type: application/json

{
  "@type": "CheckPermission",
  "permissionSpecification": {
    "@type": "PermissionSpecification",
    "function": "ProductOffering",
    "action": "create"
  },
  "user": {
    "role": "ProductManager",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "Jonathan.Goldberg",
      "@type": "PartyRoleRef",
      "@referredType": "PartyRole"
    }
  }
}
```

A more advanced scenario involves verifying whether a sales agent is authorized to apply a 10% discount to a particular product order. This type of check typically requires evaluating both the agent's permission scope and any relevant business policies associated with discounting. Through a runtime permission check, the system can determine if the sales agent meets all necessary criteria to perform the discount action on the specified order, ensuring that only properly authorized users can execute such privileged transactions.

```
POST /rolesAndPermissionsManagement/v5/checkPermission HTTP/1.1
Content-Type: application/json

{
  "@type": "CheckPermission",
  "permissionSpecification": {
    "@type": "PermissionSpecification",
    "function": "ProductOrder",
```

```
    "action": "grantAlteration"
  },
  "user": {
    "role": "SalesAgent",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "Jonathan.Goldberg",
      "@type": "PartyRoleRef",
      "@referredType": "PartyRole"
    }
  }
  "characteristic": [
    {
      "@type": "Characteristic",
      "name": "reductionPercentage",
      "value": 10
    }
  ],
  "entity": [
    {
      "@type": "EntityRef",
      "id": "P0555333313",
      "href":
"https://serverRoot/productOrderingManagement/v5/productOrder/P0555333313",
      "@referredType": "ProductOrder"
    }
  ]
}
```

Setting up Permissions for Party Roles

Permission Model Architecture

The permission model follows a hierarchical, reusable design where atomic permission specifications can be aggregated into permission specification sets, which are then assigned to party role specifications. This modular approach promotes consistency, reduces redundancy, and simplifies administration across complex organizational structures.

Permission Setup Workflow

The recommended workflow for establishing permissions follows a bottom-up approach:

- 1) Define Atomic Permissions: Create granular permission specifications that represent individual, indivisible actions on specific resources
- 2) Create Permission Sets: Aggregate related permission specifications into logical collections that represent coherent capabilities
- 3) Assign to Roles: Map permission specification sets to party role specifications, establishing the business context and meaning of these combined permissions

Practical Example: Product Catalog Management

Consider the scenario of managing permissions for product catalog operations. The business requirements include:

- Standard marketers: Create and edit product offerings and pricing, with read-only access to product specifications
- Senior marketers: All standard capabilities plus the ability to import data from partners
- Partner managers: Import/export capabilities with read-only catalog access, but no direct editing rights

Step 1: Atomic Permission Specifications

Permission Specification	Function	Action	Description
Manage Product Offering	ProductOffering	ReadWrite	Full management of product offerings
Manage Product Offering Price	ProductOfferingPrice	ReadWrite	Full management of pricing information
Access Product Specification	ProductSpecification	Read	Read-only access to product specifications
Access Product Offering	ProductOffering	Read	Read-only access to product offerings
Access Product Offering Price	ProductOfferingPrice	Read	Read-only access to pricing information

Permission Specification	Function	Action	Description
Import Catalog	ImportJob	ReadWrite	Initiate and manage catalog import operations
Export Catalog	ExportJob	ReadWrite	Initiate and manage catalog export operations

Step 2: Permission Specification Sets

Using these permission specifications, we now craft sets to reflect the different capabilities:

Permission Specification Set	Included Permissions	Business Purpose
Manage Offerings	Manage Product Offering, Manage Product Offering Price, Access Product Specification	Complete offering lifecycle management
Review Catalog	Access Product Offering, Access Product Offering Price, Access Product Specification	Read-only catalog access across all entities
Import Catalog	Import Catalog	Import-only capabilities for data ingestion
Exchange Catalog	Import Catalog, Export Catalog	Bidirectional data exchange with partners

Step 3: Party Role Specifications

Finally, we can assign these permission specification sets to relevant party role specifications:

Party Role Specification	Assigned Permission Sets	Business Context
Product Catalog Marketeer	Manage Offerings	Standard marketeer with full offering management but no import capabilities
Senior Product Catalog Marketeer	Manage Offerings, Import Catalog	Enhanced marketeer role with partner data import privileges
Catalog Partner Manager	Exchange Catalog, Review Catalog	Partnership role focused on data exchange without direct catalog modification

This hierarchical model ensures that permissions are composable, auditable, and aligned with business roles, while supporting both simple role-based scenarios and complex attribute-based requirements.

Sample Use Cases

This section includes a set of main use cases that can be performed with this API. Additional use cases can be generated using the operations and resources defined in this specification.

Use Case 1: Explicit Permission Assignment for TV Service Usage

Description

The central goal of this use case is to allow one person to create a permission for another, letting them access specific assets. For example, the owner of a TV service can give another user the right to use some features of the service, such as watching only children's movies, adjusting TV settings, or viewing documentaries.

Main Actors

Actor	Description
Asset Owner (Granter)	The person who holds legal rights over certain assets and has the authority to grant access or permissions to others. This individual can determine who may use, access, or manage these assets, either temporarily or permanently.
Permission Receiver (User)	This is the individual who is granted specific access rights by the asset owner. The permission receiver may use, view, or manage certain assets or functions as authorized, based on the limitations set by the granter.

Prerequisite: This API works on the principle that when a customer buys a product (linked to a particular account), any manageable asset created during product setup—such as registration in an eCare system or creating an account in a digital service—will be owned by the individual with administrator rights over the relevant customer and account. This ensures that the person with admin privileges at the time of purchase is designated as the asset owner.

Use Case Steps

1. The asset owner initiates a request to assign permission for another user to access their assets.
2. The Operator processes the permission creation request, which must include at least:
 - a. The duration that the permission remains valid
 - b. The identity of the user who will receive access
 - c. Details about which assets the user is being granted access to
 - d. The specific level of access for each asset, outlining the permitted actions and functions
3. The Operator checks that the requester has the authority to delegate rights for the listed assets, either as the owner or as someone with appropriate privileges. This verification may use a simple identifier or a more advanced authorization token.
4. The Operator then allocates the requested access level for the specified assets to the designated user in accordance with the owner's request.

Example of API Usage in the Context of the Use Case

The sample request below demonstrates how to generate a permission set that authorizes a specific user to access selected TV service features:

POST /rolesAndPermissionsManagement/v5/permissionSet HTTP/1.1
 Content-Type: application/json

```
{
  "@type": "PermissionSet",
  "creationDate": "2023-03-01T16:49:00.000Z",
  "description": "TV service usage granted by owner",
  "validFor": {
    "startDateTime": "2023-03-01T00:00:00.000Z",
    "endDateTime": "2023-04-01T00:00:00.000Z"
  },
  "user": {
    "role": "TVProductUser",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "p00000002",
      "name": "Charlie Owens",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "granter": {
    "role": "TVProductOwner",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "p00000001",
      "name": "John Owens",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "permission": [
    {
      "@type": "Permission",
      "managedAssetGroup": {
        "@type": "ListAssetGroup",
        "@baseType": "AssetGroup",
        "entity": [
          {
            "id": "S123456789",
            "@type": "EntityRef",
            "@referredType": "Service"
          }
        ]
      },
      "permissionSpecification": {
        "@type": "PermissionSpecification",
        "function": "TV.Service.Configuration",
        "action": "Read"
      }
    },
    {
      "@type": "Permission",
      "managedAssetGroup": {
        "@type": "ListAssetGroup",
        "@baseType": "AssetGroup",
        "entity": [
          {
            "id": "S123456789",
            "@type": "EntityRef",
            "@referredType": "Service"
          }
        ]
      }
    }
  ]
}
```

```
    ],  
    },  
    "permissionSpecification": {  
      "@type": "PermissionSpecification",  
      "function": "Children.Movies",  
      "action": "View"  
    }  
  }  
]  
}
```

Success Outcome

Once the API steps are completed, the user who was granted access may now utilize the specified manageable assets as permitted by the level of access they've received. This means the individual can interact with the authorized functions and features within the boundaries set by the permission record.

Use Case 2: Implicit Permission Assignment for Service Administration

Description

The primary goal of this use case is to generate a new set of permission specifications, establishing the authorized access levels for particular manageable assets for individuals assigned to this role.

Main Actors

Actor Name	Description
Administrator	Creates permission specification sets (user roles) that determine access rights for other users.
Asset Owner	Grants access to assets by authorizing permission specification sets for roles and users.
User	Is assigned a role, which defines what actions and access are allowed on the managed assets.

Prerequisite: This API works on the principle that when a customer buys a product (linked to a particular account), any manageable asset created during product setup—such as registration in an eCare system or creating an account in a digital service—will be owned by the individual with administrator rights over the relevant customer and account. This ensures that the person with admin privileges at the time of purchase is designated as the asset owner.

Use Case Steps

1. The admin operator submits a request to create a new permission specification set with at least the following details:
 - a. The defined access level over specific functions available on an asset and the permitted actions on those functions.
2. Once the permission specification set is established, the asset owner (or admin) submits a request to assign a permission to another user, referencing the set and including at least:
 - a. The validity period of the permission
 - b. The user who will receive access
 - c. Information regarding the manageable assets the user will access
 - d. The permission specification set applicable to the user and referenced asset
3. The Operator verifies that the requester is authorized to assign permissions on the referenced assets (for example, is the asset owner or otherwise has appropriate access). This authorization can be based on the requester's identifier or a token-based approach.
4. The Operator then allocates the requested access level to the individual for the specified manageable assets, as authorized by the asset owner.

Example of API Usage in the Context of the Use Case

- The following example demonstrates how a service admin permission specification set can be created:

REQUEST

```
POST /rolesAndPermissionsManagement/v5/permissionSpecificationSet HTTP/1.1
Content-Type: application/json
```

```
{
  "@type": "PermissionSpecificationSet",
  "involvementRole": "Service.Admin",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecification",
      "name": "service:readwrite",
      "function": "Service",
      "action": "ReadWrite"
    }
  ]
}
```

RESPONSE

```
200
Content-Type: application/json
```

```
{
  "id": "8a2a86e7-1ab2-4bb6-b447-a6d2e14b3d74",
  "@type": "PermissionSpecificationSet",
  "involvementRole": "Service.Administrator",
  "permissionSpecification": [
    {
      "id": "17f3619f-6b66-419d-b672-83ba5facd98a",
      "@type": "PermissionSpecification",
      "name": "service:readwrite",
      "function": "Service",
      "action": "ReadWrite"
    }
  ]
}
```

- The asset's owner utilizes the Operator's service to generate a new permission entry. This entry assigns a specific role to a user for a selected asset.
- Lastly, the example request below demonstrates how to allocate the previously constructed permission specification set to an individual user on behalf of the owner.

REQUEST

```
POST /rolesAndPermissionsManagement/v5/permissionSet HTTP/1.1
Content-Type: application/json
```

```
{
```

```

    "@type": "PermissionSet",
    "creationDate": "2023-03-01T16:49:00.000Z",
    "description": "TV service usage granted by owner",
    "validFor": {
      "startDateTime": "2023-03-01T00:00:00.000Z",
      "endDateTime": "2023-04-01T00:00:00.000Z"
    },
    "user": {
      "role": "TVProductUser",
      "@type": "RelatedSecurityPrincipal",
      "securityPrincipal": {
        "id": "p00000003",
        "name": "Linda Owens",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    },
    "granter": {
      "role": "TVProductOwner",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "p00000001",
        "name": "John Owens",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    },
    "permission": [
      {
        "@type": "Permission",
        "managedAssetGroup": {
          "@type": "ListAssetGroup",
          "@baseType": "AssetGroup",
          "entity": [
            {
              "id": "S123456789",
              "@type": "EntityRef",
              "@referredType": "Service"
            }
          ]
        },
        "permissionSpecificationSet": {
          "@type": "PermissionSpecificationSetRef",
          "id": "8a2a86e7-1ab2-4bb6-b447-a6d2e14b3d7"
        }
      }
    ]
  }
}

```

Success Outcome

Once these API operations are finished, the authorized individual is able to interact with the specified manageable assets according to the level of access established by the assigned permission specification set.

Use Case 3: Legal Guardian on Behalf of Protected Adult

Description

The primary objective of this scenario is to grant permissions to a Legal Guardian, enabling her to manage the Products owned by a Customer who is classified as a Protected Adult.

NOTE

This use case is documented within the ODA Implementation Guidelines for use case realization, available here: <https://projects.tmforum.org/wiki/display/PUB/IG1228+Use+Case+UC006:+Legal+guardian+v4>.

For comprehensive details, refer to that resource. This section focuses only on the aspects related to Roles and Permissions.

Actor	Description
CSR (Customer Service Representative)	A staff member responsible for setting up and granting the appropriate access rights or privileges to the guardian, ensuring they can act on behalf of another individual within the system.
Guardian	The designated individual who has been given specific rights by the CSR. The guardian is entrusted to manage, oversee, or act on behalf of another person, often in support or supervisory roles.

Prerequisite: Within this use case, the guardian is already registered as a customer with the CSP. However, such registration is not a general prerequisite for guardianship.

Use Case Steps

1. The guardian contacts the Customer Service Representative (CSR) and submits the required proof (such as court documentation).
2. The CSR grants the guardian Guardianship permissions, establishing validity based on the duration specified by the provided proof.
3. The guardian signs in to the self-service portal, where they can view both the products owned by the protected adult and their own products, due to the assigned Guardianship permissions.

Example of API Usage in the Context of the Use Case

The following API operations facilitate this use case:

- The Customer Service Representative (CSR) performs an operation to assign permissions.
- The guardian executes an operation to verify permissions on the protected adult's products.

Success Outcome

After these API operations are completed, the authorized individual can utilize the designated manageable assets in accordance with the access level specified by the assigned permission specification set.

Use Case 4: Implicit Permission Assignment for Customer Selfcare Administration

Description

The primary goal of this use case is to establish a customer self-care administrator role (permission specification set). This role is endowed with specific entitlements, allowing the holder to perform authorized actions on a designated set of manageable assets.

Main Actors

Actor	Description
Admin Operator	The system administrator responsible for creating and managing permission specification sets, commonly called user roles . These roles define groups of allowed actions that can later be assigned to individuals.
Asset Owner (Granter)	The person or entity who holds authority over specific assets (such as data, resources, or services). This actor has the power to grant or delegate access rights to others.
User (Role Holder)	The individual who is assigned a specific role. By receiving this role, the user inherits the permissions and responsibilities defined by the Admin Operator and authorized by the Asset Owner.

Prerequisite: This API works on the principle that when a customer buys a product (linked to a particular account), any manageable asset created during product setup—such as registration in an eCare system or creating an account in a digital service—will be owned by the individual with administrator rights over the relevant customer and account. This ensures that the person with admin privileges at the time of purchase is designated as the asset owner.

Use Case Steps

1. The admin operator sends a request to create a new permission specification set, providing at least the following details:
 - a. The access level granted over a set of asset functions and the permitted actions on those functions
2. Once defined, the asset owner or admin submits a request to assign this permission to another user, supplying the following minimum information:
 - a. The validity period of the permission
 - b. The user being granted access
 - c. The manageable assets covered by the permission
 - d. The specific permission specification set to be allocated
3. The Operator verifies that the requester is authorized to grant permissions for the specified manageable assets (i.e., is either the owner or possesses sufficient access level). This can be verified via the requester's ID or by more advanced token-based authorization methods.
4. The Operator then allocates the requested access to the manageable assets for the

designated individual, as authorized by the owner.

Example of API Usage in the Context of the Use Case

- The following example request and response demonstrate how to create a service admin permission specification set:

REQUEST

```
POST /rolesAndPermissionsManagement/v5/permissionSpecificationSet HTTP/1.1
Content-Type: application/json

{
  "@type": "PermissionSpecificationSet",
  "involvementRole": "Customer.Selfcare.Admin",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecification",
      "description": "Permission to manage users and to add or remove permissions to delegated users.",
      "name": "usermanagement:readwrite",
      "function": "user_management",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "description": "Permission to view and change account data details.",
      "name": "accountdetails:readwrite",
      "function": "account_details",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "description": "Permission to place request to change holder data.",
      "name": "holderdata:change",
      "function": "change_holder_data",
      "action": "ReadWrite",
      "specificationCharacteristic": [
        {
          "@type": "CharacteristicSpecification",
          "name": "Macro.Segment",
          "valueType": "string",
          "characteristicValueSpecification": [
            {
              "@type": "StringCharacteristicValueSpecification",
              "valueType": "string",
              "value": "mass_market"
            },
            {
              "@type": "StringCharacteristicValueSpecification",
              "valueType": "string",
              "value": "residential"
            }
          ]
        }
      ]
    }
  ]
}
```

RESPONSE

```

200
Content-Type: application/json

{
  "@type": "PermissionSpecificationSet",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
  edd9d8a4",
  "id": "pss-edd9d8a4",
  "involvementRole": "Customer.Selfcare.Admin",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecification",
      "href":
      "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
      dd798b34",
      "id": "ps-dd798b34",
      "description": "Permission to manage users and to add or remove permissions to
      delegated users.",
      "name": "usermanagement:readwrite",
      "function": "user_management",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "href":
      "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
      591c77a2",
      "id": "ps-591c77a2",
      "description": "Permission to view and change account data details.",
      "name": "accountdetails:readwrite",
      "function": "account_details",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "href":
      "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
      9f32c9e0",
      "id": "ps-9f32c9e0",
      "description": "Permission to place request to change holder data.",
      "name": "holderdata:change",
      "function": "change_holder_data",
      "action": "ReadWrite",
      "specificationCharacteristic": [
        {
          "@type": "CharacteristicSpecification",
          "name": "Macro.Segment",
          "valueType": "string",
          "characteristicValueSpecification": [
            {
              "@type": "StringCharacteristicValueSpecification",
              "valueType": "string",
              "value": "mass_market"
            },
            {
              "@type": "StringCharacteristicValueSpecification",
              "valueType": "string",
              "value": "residential"
            }
          ]
        }
      ]
    }
  ]
}

```

```

    }
  ]
}
]
}
}

```

- The owner of the manageable assets uses the service provided by the Operator to create a new permission record, which assigns a specific role to a user for a selected asset.
- Finally, the following request example can be used to assign the previously created permission specification set to an individual user from its owner.

REQUEST

```

POST /rolesAndPermissionsManagement/v5/permissionSet HTTP/1.1
Content-Type: application/json

{
  "@type": "PermissionSet",
  "description": "Customer selfcare admin profile granted by owner to delegate
management of customers, products and services entities.",
  "validFor": {
    "startDateTime": "2023-03-01T00:00:00.000Z",
    "endDateTime": "2023-04-01T00:00:00.000Z"
  },
  "user": {
    "role": "ProductUser",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "p00000003",
      "name": "Linda Owens",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "granter": {
    "role": "ProductOwner",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "p00000001",
      "name": "John Owens",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "permission": [
    {
      "@type": "Permission",
      "managedAssetGroup": {
        "@type": "ListAssetGroup",
        "@baseType": "AssetGroup",
        "entity": [
          {
            "@type": "EntityRef",
            "id": "C123456789",
            "@referredType": "Customer"
          }
        ]
      }
    }
  ]
}

```

```

    }
  ],
  "permissionSpecificationSet": {
    "@type": "PermissionSpecificationSet",
    "id": "pss-edd9d8a4"
  }
},
{
  "@type": "Permission",
  "managedAssetGroup": {
    "@type": "ListAssetGroup",
    "@baseType": "AssetGroup",
    "entity": [
      {
        "@type": "EntityRef",
        "id": "5.99999.99.99.999999",
        "@referredType": "Customer"
      }
    ]
  },
  "permissionSpecificationSet": {
    "@type": "PermissionSpecificationSet",
    "id": "pss-edd9d8a4"
  }
},
{
  "@type": "Permission",
  "managedAssetGroup": {
    "@type": "ListAssetGroup",
    "@baseType": "AssetGroup",
    "entity": [
      {
        "@type": "EntityRef",
        "id": "1-44up07",
        "@referredType": "Product"
      }
    ]
  },
  "permissionSpecificationSet": {
    "@type": "PermissionSpecificationSet",
    "id": "pss-edd9d8a4"
  }
},
{
  "@type": "Permission",
  "managedAssetGroup": {
    "@type": "ListAssetGroup",
    "@baseType": "AssetGroup",
    "entity": [
      {
        "@type": "EntityRef",
        "id": "S123456789",
        "@referredType": "Service"
      }
    ]
  },
  "permissionSpecificationSet": {
    "@type": "PermissionSpecificationSet",
    "id": "pss-edd9d8a4"
  }
}
]

```

```
}
```

Success Outcome

After these API operations are completed, the user granted access can utilize the designated manageable assets in alignment with the access level specified by the assigned permission specification set.

Use Case 5: Resource Role-Based Access Control Configuration for a TMF Open API

This use case describes the process by which an administrative user utilizes a Resource Catalog Management application to establish role-based access control between an API consumer (Product Catalog Management Web Application) and an API provider (TMF620 Product Catalog Management API) in accordance with TM Forum Open API and Information Framework (SID) specifications.

Description

An administrative user accesses the Resource Catalog Management application to configure access between the Product Catalog Management Web Application and the TMF620 Product Catalog Management API. The process consists of the following main steps:

- The administrator defines a resource role labelled "API Consumer" and assigns it to the Product Catalog Management Web Application. This role encapsulates the permissions required for the web application to interact with protected resources exposed via the TMF620 API, such as retrieval and querying of product catalog entities, consistent with the API's defined access control policies.
- Concurrently, the administrator defines a resource role labelled "API Provider" and assigns it to the TMF620 Product Catalog Management API. This role declares the full set of permissions, supported operations (e.g., GET, POST, PATCH, DELETE), and entitlement scopes that the API exposes to authorized consumers.

Through this configuration, a bilateral relationship is established: the Product Catalog Management Web Application, as an API Consumer, receives specific entitlements to interact with the TMF620 API, while the TMF620 API, as an API Provider, exposes its capabilities and access policies in accordance with TM Forum's role-based access control model.

Outcome

Upon completion, the administrative configuration ensures secure, standardized interoperability between consumer and provider systems:

- The API Consumer role enables the Product Catalog Management Web Application to access designated resources through the TMF620 API.
- The API Provider role defines and governs which operations and resources are available to authorized consumers.

This role-based access control model is fully aligned with TM Forum Open API architecture and the Information Framework (SID), supporting scalable, policy-driven integration and management of digital services.

Figure 5 illustrates the resulting configuration, highlighting the Product Catalog Management Web Application as an API Consumer interacting with the TMF620 Product Catalog API as an API Provider.

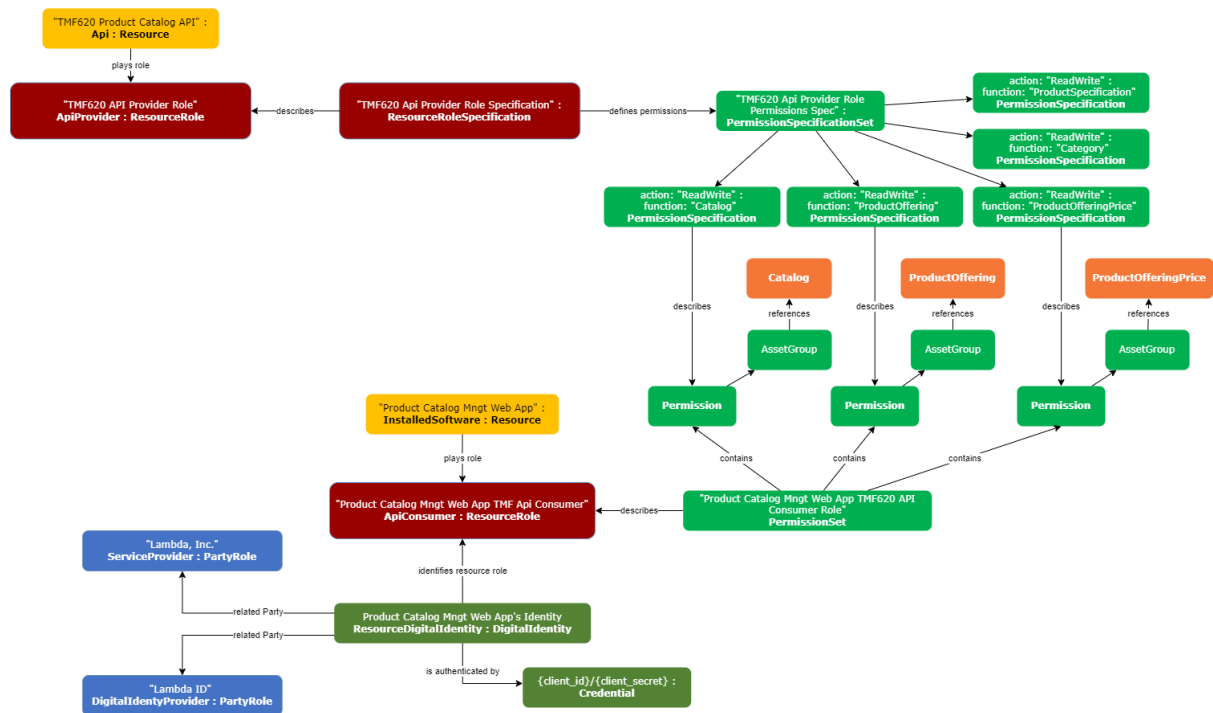


Figure 5. Product Catalog Management Web App Api Consumer example

Additional Use Cases (RBAC)

- On-boarding CSP Employee as Junior CSR user
- Junior CSR Employee performs customer maintenance
- Self-service user has access to her own customer maintenance activities

Additional Use Cases (ABAC)

- Consumer CSR can only access Consumer customer accounts and B2B CSR can access only B2B customer
- Consumer CSR in North market can access customers, products, etc. only in that market
- Account Managers could access only the customers which they manage - e.g., PepsiCo account manager cannot see Coca Cola details
- Enterprise Customer gives rights to certain employees to manage specific parts of the structure, e.g., Joe Doe can manage all the mobiles held in legal department, Jane Dane can manage the mobiles of sales, etc.
- Network Department supervisor gives rights to network engineer to manage some NFVs
- Catalog Administrator gives rights to marketer to create offerings in Mobile line-of-business
- Customers are enabled to self-provision some service characteristics

Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema-based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a `FilterAssetGroup` and `ListAssetGroup` inheriting properties from the abstract `AssetGroup` entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines v3.0 Part 2 document.

The `@type` attribute provides a way to represent the actual class type of an entity. For example, within a list of `AssetGroup` instances some may be instances of `FilterAssetGroup` where other could be instances of `ListAssetGroup`. The `@type` gives this information. All resources and sub-resources of this API have a `@type` attributes that can be provided when this is useful.

The `@referredType` can be used within reference entities (like for instance an `AssetGroupRef` object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the `@type`, when used, denotes the class type of the reference itself, such as `FilterAssetGroupRef` or `ListAssetGroupRef`, and not the class type of the referred object. However, since reference classes are rarely sub-classed, `@type` is generally not useful in reference objects.

The `@schemaLocation` property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected *characteristics* of an entity.

The `@baseType` attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

RESOURCE MODEL

Managed Entity and Task Resource Models

PermissionSet resource

The PermissionSet is a set of Permissions granted to a user (party, party role or resource). The set may be granted explicitly by an authorized user or may be acquired implicitly due to the role that the user is playing.

Resource model

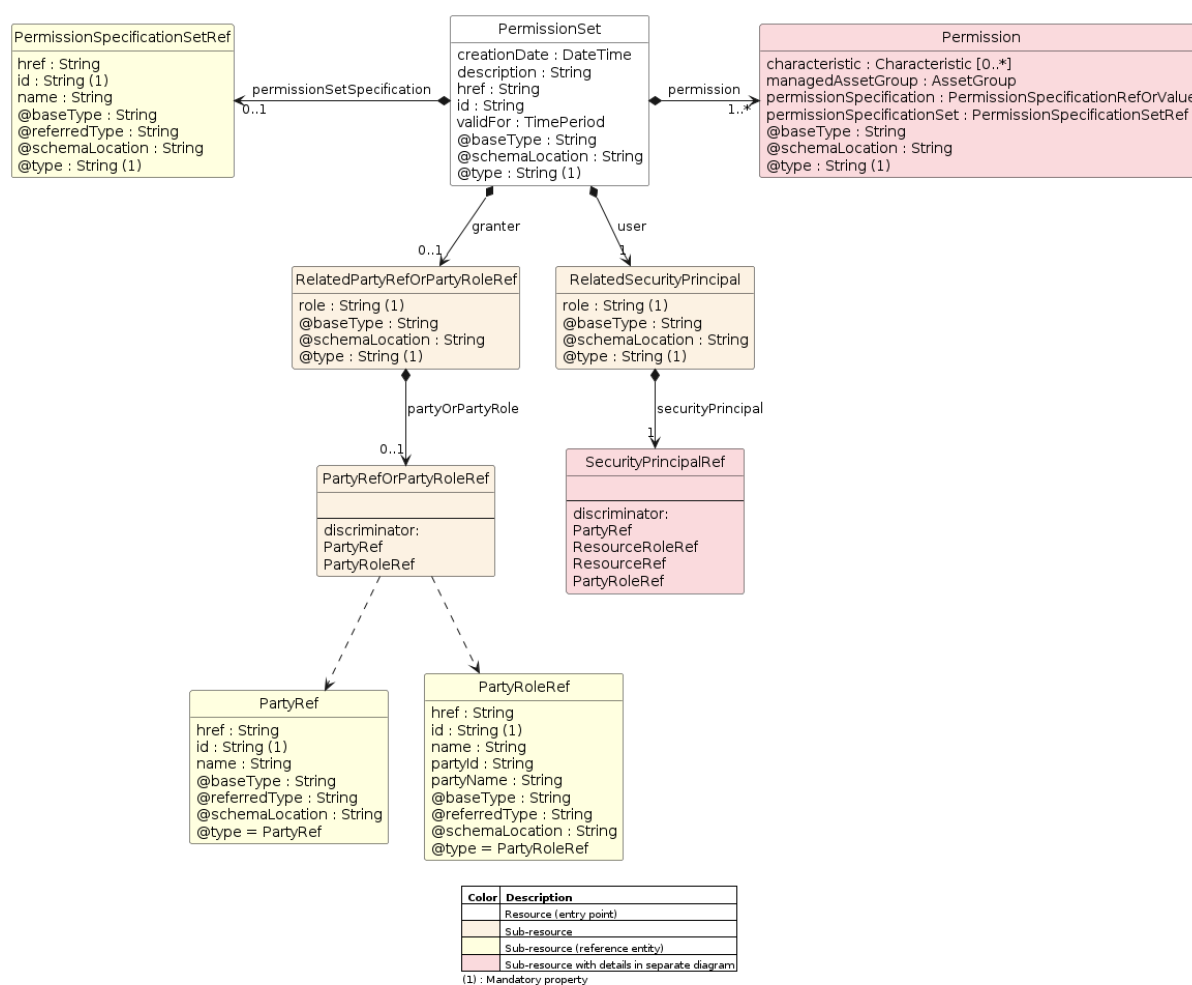


Figure 1 - PermissionSet

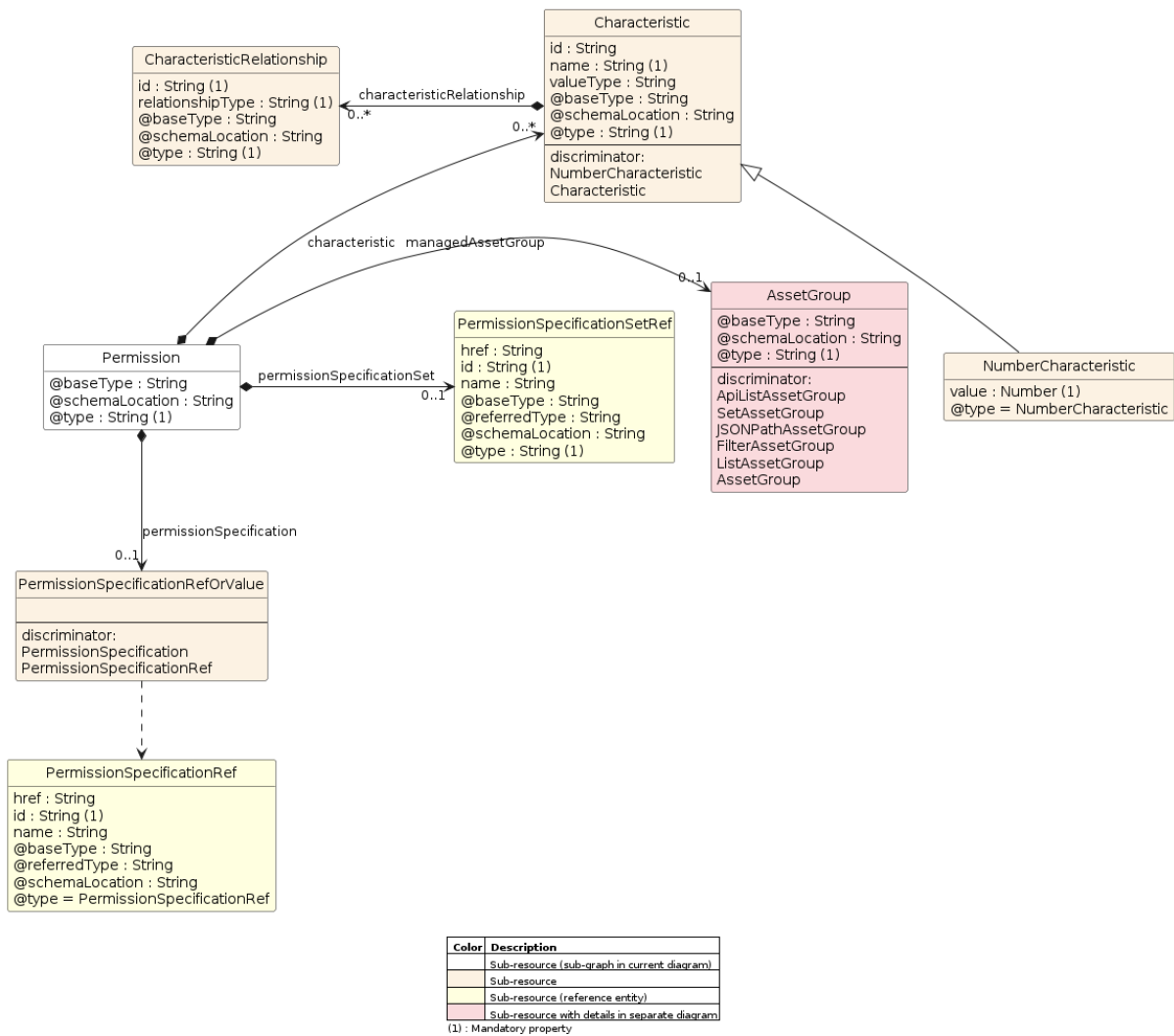


Figure 2 - Permission

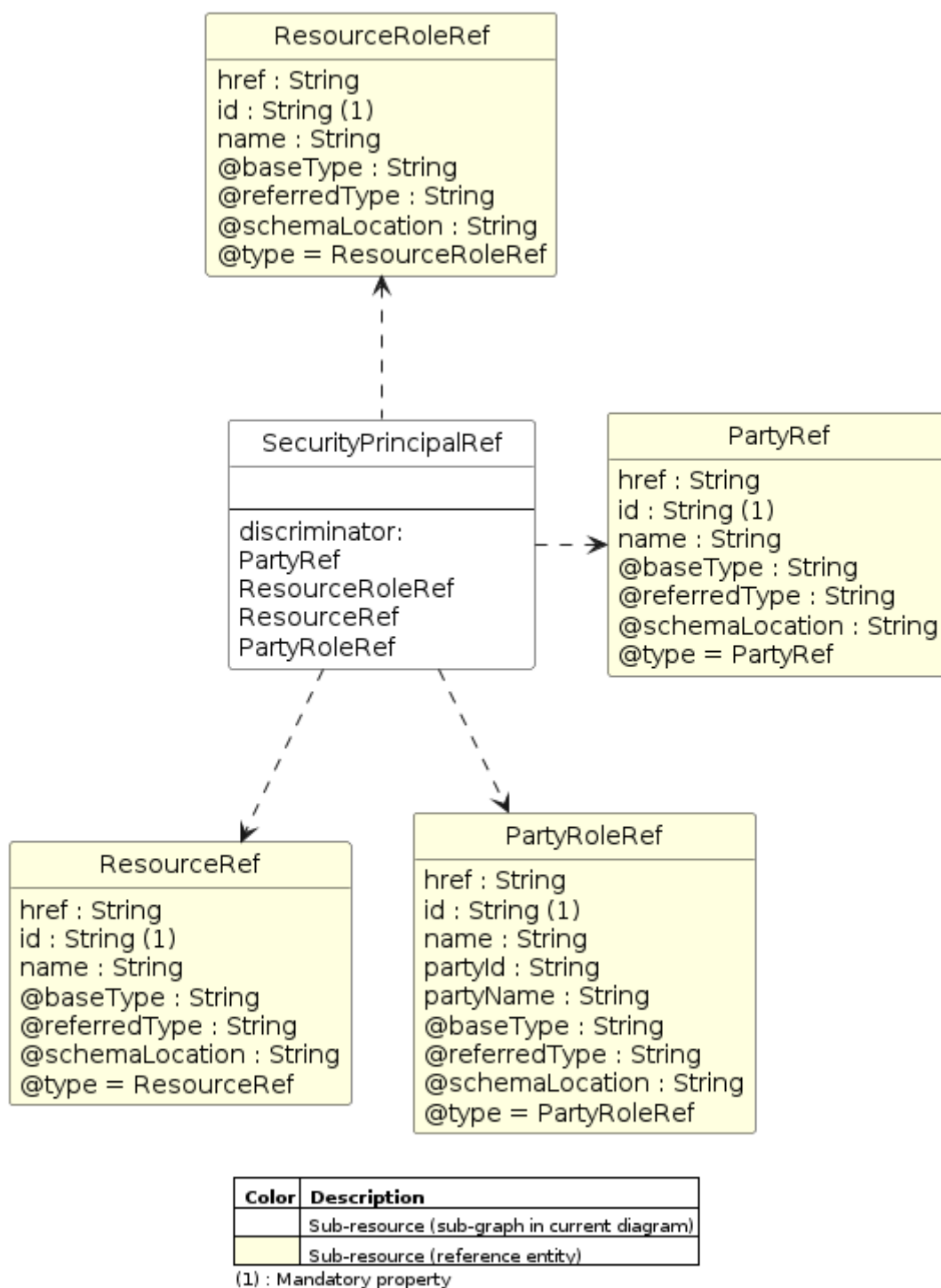


Figure 3 - SecurityPrincipalRef

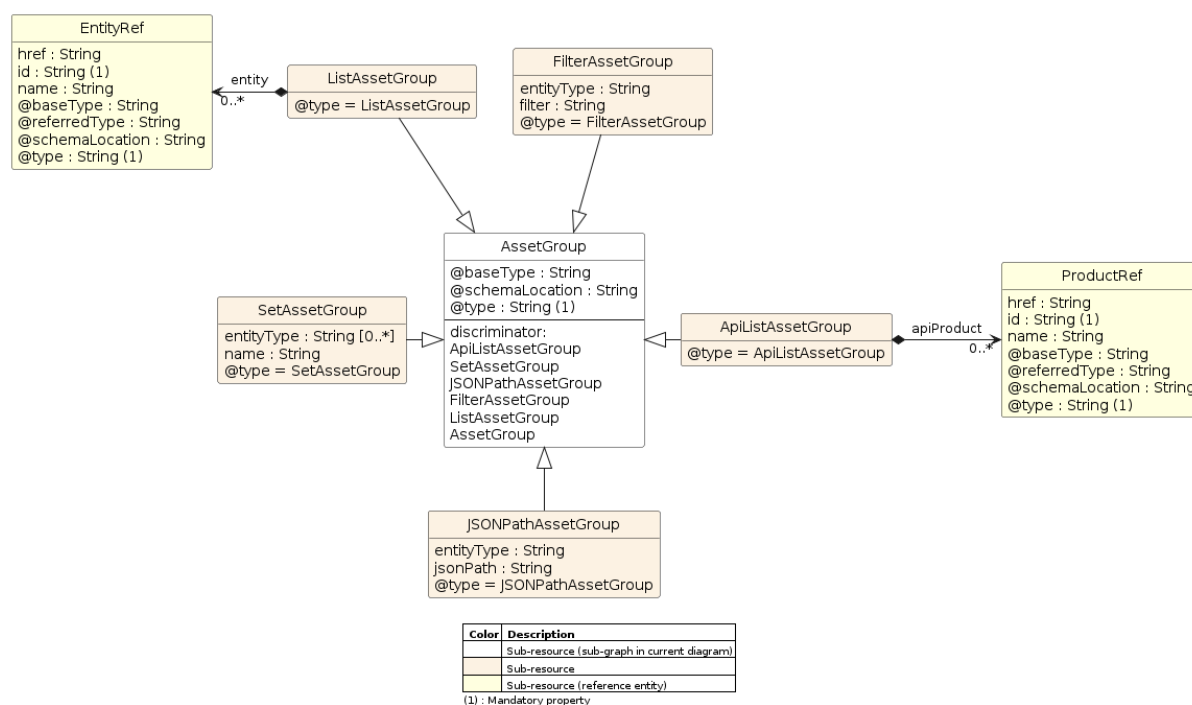


Figure 4 - AssetGroup

Field descriptions

PermissionSet fields

creationDate	A DateTime. Date when the PermissionSet was created and assigned to a user.
description	A String. Text describing the permission set.
granter	A RelatedPartyRefOrPartyRoleRef. RelatedParty reference. A related party defines party or party role or its reference, linked to a specific entity.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
permission	A Permission. The list of permissions in this set.
permissionSetSpecification	A PermissionSpecificationSetRef. A PermissionSpecificationSet defines a set of permissions that can be attached to a party role specification. The resulting permissions will then apply to all users that adopt this party role.
user	A RelatedSecurityPrincipal. A RelatedSecurityPrincipal defines a SecurityPrincipal reference, linked to a specific entity.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

@type	A String. When sub-classing, this defines the sub-class Extensible name.
-------	--

RelatedPartyRefOrPartyRoleRef sub-resource fields

partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

@type	A String. When sub-classing, this defines the sub-class Extensible name.
-------	--

Permission sub-resource fields

characteristic	A Characteristic. List of characteristics that apply to this permission, according to the corresponding specification. Characteristic can be instantiated as * NumberCharacteristic
managedAssetGroup	An AssetGroup. Defines a group of assets (entities) to which access can be controlled. Specific subclasses allow definition in various ways (list of entities, queries, ownership by customer, etc.). AssetGroup is not a managed entity, its concrete subclasses are always included by value in their containing entities. AssetGroup can be instantiated as * ApiListAssetGroup * FilterAssetGroup * JSONPathAssetGroup * ListAssetGroup * SetAssetGroup
permissionSpecification	A PermissionSpecificationRefOrValue. Related Entity reference. A related PermissionSpecification defines a PermissionSpecification described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the PermissionSpecification entity and not the PermissionSpecificationRefOrValue class itself.
permissionSpecificationSet	A PermissionSpecificationSetRef. A PermissionSpecificationSet defines a set of permissions that can be attached to a party role specification. The resulting permissions will then apply to all users that adopt this party role.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.

@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	Characteristic can be instantiated as * NumberCharacteristic
value	This property is present in subclasses

CharacteristicRelationship sub-resource fields

id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship. Collection of characteristic relationships.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AssetGroup sub-resource fields

@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

	AssetGroup can be instantiated as * ApiListAssetGroup * FilterAssetGroup * JSONPathAssetGroup * ListAssetGroup * SetAssetGroup
apiProduct	This property is present in subclasses
entity	This property is present in subclasses
entityType	This property is present in subclasses
filter	This property is present in subclasses
jsonPath	This property is present in subclasses
name	This property is present in subclasses

ProductRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

EntityRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ApiListAssetGroup sub-resource fields

apiProduct	A ProductRef. The list of products for which permissions are given.
------------	---

@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FilterAssetGroup sub-resource fields

entityType	A String. The type of entity in the group, equivalent to @type in the entity definition. Must be a top-level managed entity. The filter expression is applied to this entity type.
filter	A String. The filter expression that defines the entities covered by a permission set, format of the filter is as per the simple query parameter filter as defined in TMF630.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

JSONPathAssetGroup sub-resource fields

entityType	A String. The type of entity in the group, equivalent to @type in the entity definition. Must be a top-level managed entity. The JSON Path expression is applied to this entity type.
jsonPath	A String. The JSON Path expression that defines the entities covered by a permission set, format of the filter is as per the JSON Path query parameter filter as defined in TMF630.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ListAssetGroup sub-resource fields

entity	An EntityRef. List of entities to which permissions might apply.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

SetAssetGroup sub-resource fields

entityType	A String. List of entity types for this named list.
name	A String. Name of the list.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PermissionSpecification sub-resource fields

action	A String. Level of access granted as part of the permission, such as read or write.
description	A String. A fuller description of the permission specification, should describe to a human user what the purpose is of this permission.
function	A String. The function to which the action applies, could be an entity class name, in which case the action defines permission to read or modify these entities. E.g. Read ProductOffering. Alternatively, the function is an agreed string that reflects a specific functionality to which access is controlled, such as giving an ad-hoc price reduction in a product order.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A descriptive name of the permission specification, should summarize to a human user what the purpose is of this permission.
specificationCharacteristic	A CharacteristicSpecification. List of characteristics that are relevant for evaluating the permission. For example, if the permission relates to granting an ad-hoc order price reduction, a characteristic might be defined for the maximum allowed reduction.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicSpecification sub-resource fields

charSpecRelationship	A CharacteristicSpecificationRelationship. An aggregation, migration, substitution, dependency or exclusivity relationship between/among Specification Characteristics.
----------------------	---

characteristicValueSpecification	A CharacteristicValueSpecification. A CharacteristicValueSpecification object is used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a CharacteristicSpecification object. The values of the attributes in the CharacteristicValueSpecification object describe the values of the attributes that a corresponding Characteristic object can take on. CharacteristicValueSpecification can be instantiated as * BooleanArrayCharacteristicValueSpecification * BooleanCharacteristicValueSpecification * FloatArrayCharacteristicValueSpecification * FloatCharacteristicValueSpecification * IntegerArrayCharacteristicValueSpecification * IntegerCharacteristicValueSpecification * MapArrayCharacteristicValueSpecification * MapCharacteristicValueSpecification * NumberArrayCharacteristicValueSpecification * NumberCharacteristicValueSpecification * ObjectArrayCharacteristicValueSpecification * ObjectCharacteristicValueSpecification * StringArrayCharacteristicValueSpecification * StringCharacteristicValueSpecification
configurable	A Boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A String. A narrative that explains the CharacteristicSpecification.
extensible	A Boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A String. Unique ID for the characteristic.
isUnique	A Boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristic. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An Integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An Integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A String. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A String. A rule or principle represented in regular expression used to derive the value of a characteristic value.

validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
valueType	A String. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
@valueSchemaLocation	A String. This (optional) field provides a link to the schema describing the value type.

CharacteristicSpecificationRelationship sub-resource fields

characteristicSpecificationId	A String. Unique identifier of the characteristic within the specification.
name	A String. Name of the target characteristic within the specification.
parentSpecificationHref	A Uri. Hyperlink reference to the parent specification containing the target characteristic.
parentSpecificationId	A String. Unique identifier of the parent specification containing the target characteristic.
relationshipType	A String. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.

validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
	CharacteristicValueSpecification can be instantiated as * BooleanArrayCharacteristicValueSpecification * BooleanCharacteristicValueSpecification * FloatArrayCharacteristicValueSpecification * FloatCharacteristicValueSpecification * IntegerArrayCharacteristicValueSpecification * IntegerCharacteristicValueSpecification * MapArrayCharacteristicValueSpecification * MapCharacteristicValueSpecification * NumberArrayCharacteristicValueSpecification * NumberCharacteristicValueSpecification * ObjectArrayCharacteristicValueSpecification * ObjectCharacteristicValueSpecification * StringArrayCharacteristicValueSpecification * StringCharacteristicValueSpecification
value	This property is present in subclasses

BooleanArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Boolean.
valueFrom	An Integer. The low range value that a characteristic can take on.

valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Boolean. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".

regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Float.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Float. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.

@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	An Integer.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.

validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	An Integer. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

MapArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	An object.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

MapCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	An object. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Number.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.

valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A Number. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	An object. Collection of characteristic values.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	An object. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

@type	A String. When sub-classing, this defines the sub-class Extensible name.
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StringArrayCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A String.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

value	A String. Value of the characteristic.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PermissionSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PermissionSpecificationSetRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PermissionSpecificationSetRef sub-resource fields

See [\[PermissionSpecificationSetRef\]](#) for details.

RelatedSecurityPrincipal sub-resource fields

role	A String. Role played by the related security principal in the context of the specific entity it is linked to.
securityPrincipal	A SecurityPrincipalRef. A reference to a security principal.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'PermissionSet' resource object.

```
{
  "@type": "PermissionSet",
  "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSet/Prms-jnzgh456",
  "id": "Prms-jnzgh456",
```

```

"creationDate": "2019-10-01T00:00:00.000Z",
"description": "superProfile Granted for user",
"validFor": {
  "startDateTime": "2019-10-01T00:00:00.000Z",
  "endDateTime": "2019-10-31T23:59:59.000Z"
},
"user": {
  "role": "client",
  "@type": "RelatedSecurityPrincipal",
  "securityPrincipal": {
    "id": "cust-745712-A",
    "href": "https://host:port/partyManagement/v5/party/cust-745712-A",
    "name": "Jack Marshall",
    "@type": "PartyRef",
    "@referredType": "Individual"
  }
},
"granter": {
  "role": "customer advisor",
  "@type": "RelatedPartyRefOrPartyRoleRef",
  "partyOrPartyRole": {
    "id": "cclt-456745",
    "href": "https://host:port/partyManagement/v5/party/cclt-456745",
    "name": "John Baker",
    "@type": "PartyRef",
    "@referredType": "Individual"
  }
},
"permission": [
  {
    "@type": "Permission",
    "permissionSpecification": {
      "@type": "PermissionSpecification",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-31d5b770",
      "id": "ps-31d5b770",
      "name": "ProductOrder:update",
      "function": "ProductOrder",
      "action": "update"
    },
    "managedAssetGroup": {
      "@baseType": "AssetGroup",
      "@type": "JSONPathAssetGroup",
      "entityType": "ProductOrder",
      "jsonPath": "filter=productSpecification=='tv-channel'&&productCharacteristic[?(@.name=='tvrating' && @.value=='children']"
    }
  }
]
}

```

PermissionSpecificationSet resource

A PermissionSpecificationSet defines a set of permissions that can be attached to a party role specification. The resulting permissions will then apply to all users that adopt this party role.

Resource model

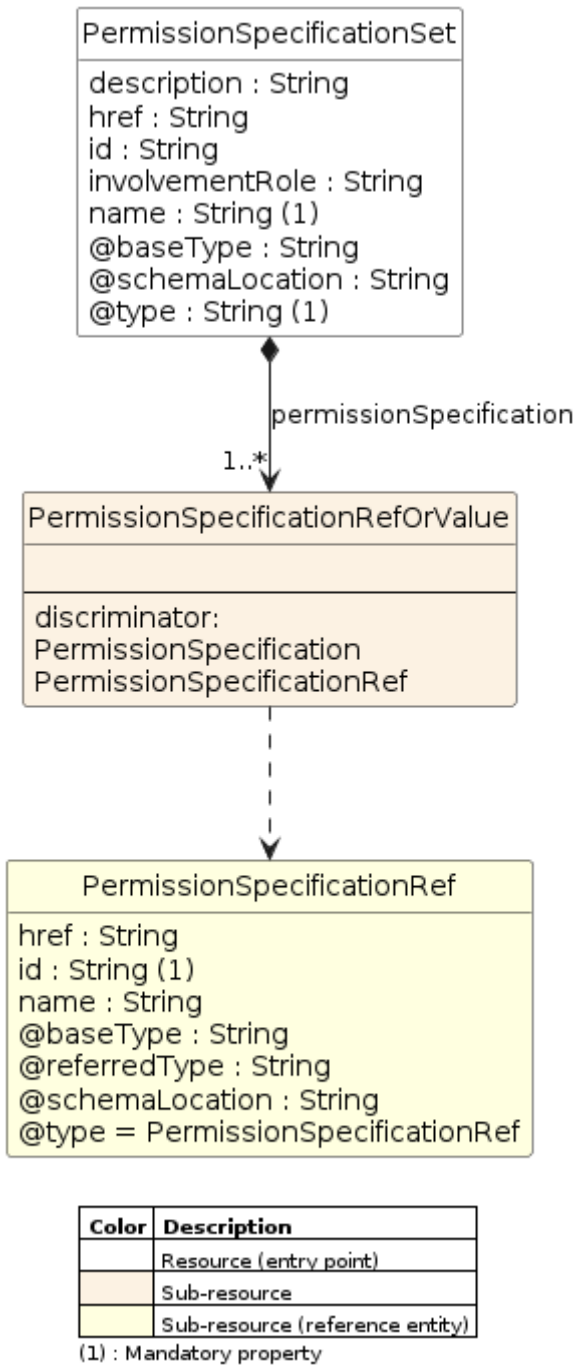


Figure 5 - PermissionSpecificationSet

Field descriptions

PermissionSpecificationSet fields

description	A String. A fuller description of the permission specification set, should describe to a human user what the purpose is of this set.
href	A String. Hyperlink reference.
id	A String. Unique identifier.

involvementRole	A String. Indication of the part that a user plays in its involvement with a manageable asset (product, service or resource).
name	A String. A descriptive name of the permission specification set, should summarize to a human user what the purpose is of this set.
permissionSpecification	A PermissionSpecificationRefOrValue. List of permission specifications in this set.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PermissionSpecification sub-resource fields

See [\[PermissionSpecification\]](#) for details.

CharacteristicSpecification sub-resource fields

See [\[CharacteristicSpecification\]](#) for details.

CharacteristicSpecificationRelationship sub-resource fields

See [\[CharacteristicSpecificationRelationship\]](#) for details.

CharacteristicValueSpecification sub-resource fields

See [\[CharacteristicValueSpecification\]](#) for details.

BooleanArrayCharacteristicValueSpecification sub-resource fields

See [\[BooleanArrayCharacteristicValueSpecification\]](#) for details.

BooleanCharacteristicValueSpecification sub-resource fields

See [\[BooleanCharacteristicValueSpecification\]](#) for details.

FloatArrayCharacteristicValueSpecification sub-resource fields

See [\[FloatArrayCharacteristicValueSpecification\]](#) for details.

FloatCharacteristicValueSpecification sub-resource fields

See [\[FloatCharacteristicValueSpecification\]](#) for details.

IntegerArrayCharacteristicValueSpecification sub-resource fields

See [\[IntegerArrayCharacteristicValueSpecification\]](#) for details.

IntegerCharacteristicValueSpecification sub-resource fields

See [\[IntegerCharacteristicValueSpecification\]](#) for details.

MapArrayCharacteristicValueSpecification sub-resource fields

See [\[MapArrayCharacteristicValueSpecification\]](#) for details.

MapCharacteristicValueSpecification sub-resource fields

See [\[MapCharacteristicValueSpecification\]](#) for details.

NumberArrayCharacteristicValueSpecification sub-resource fields

See [\[NumberArrayCharacteristicValueSpecification\]](#) for details.

NumberCharacteristicValueSpecification sub-resource fields

See [\[NumberCharacteristicValueSpecification\]](#) for details.

ObjectArrayCharacteristicValueSpecification sub-resource fields

See [\[ObjectArrayCharacteristicValueSpecification\]](#) for details.

ObjectCharacteristicValueSpecification sub-resource fields

See [\[ObjectCharacteristicValueSpecification\]](#) for details.

StringArrayCharacteristicValueSpecification sub-resource fields

See [\[StringArrayCharacteristicValueSpecification\]](#) for details.

StringCharacteristicValueSpecification sub-resource fields

See [\[StringCharacteristicValueSpecification\]](#) for details.

PermissionSpecificationRef sub-resource fields

See [\[PermissionSpecificationRef\]](#) for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'PermissionSpecificationSet' resource object.

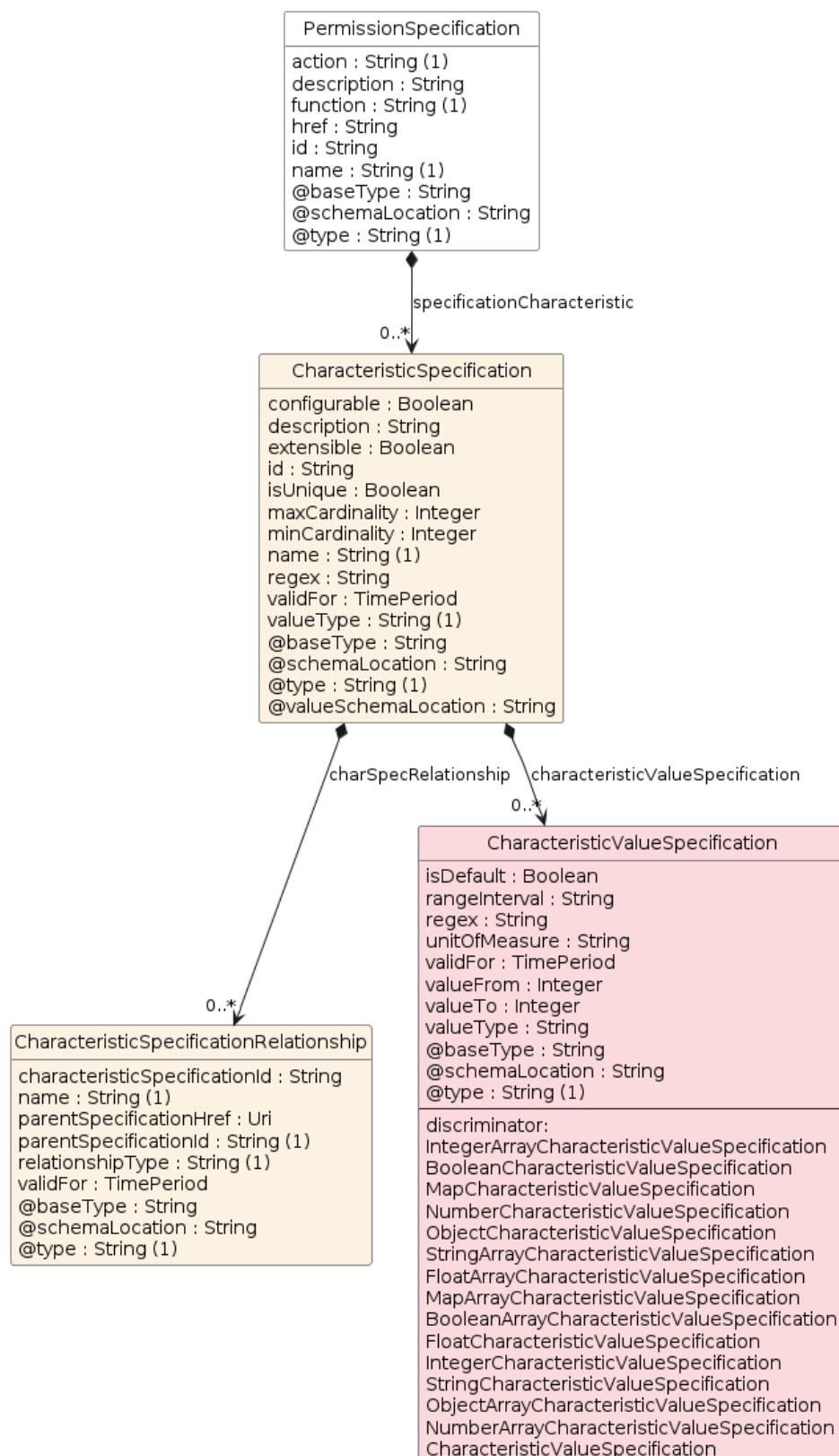
```
{
  "@type": "PermissionSpecificationSet",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
  ebe41def",
  "id": "pss-ebe41def",
  "involvementRole": "MyTV familyAdmin Profile",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecificationRef",
      "@referredType": "PermissionSpecification",
      "href":
```

```
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-5afe52c5",
  "id": "pss-5afe52c5"
},
{
  "@type": "PermissionSpecificationRef",
  "@referredType": "PermissionSpecification",
  "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-769ef409",
  "id": "pss-769ef409"
}
]
```

PermissionSpecification resource

A PermissionSpecification defines a permissible action (e.g. read, write) on some function, where a function can be an entity class name or some specific functionality.

Resource model



Color	Description
	Resource (entry point)
	Sub-resource
	Sub-resource with details in separate diagram

(1) : Mandatory property

Figure 6 - PermissionSpecification

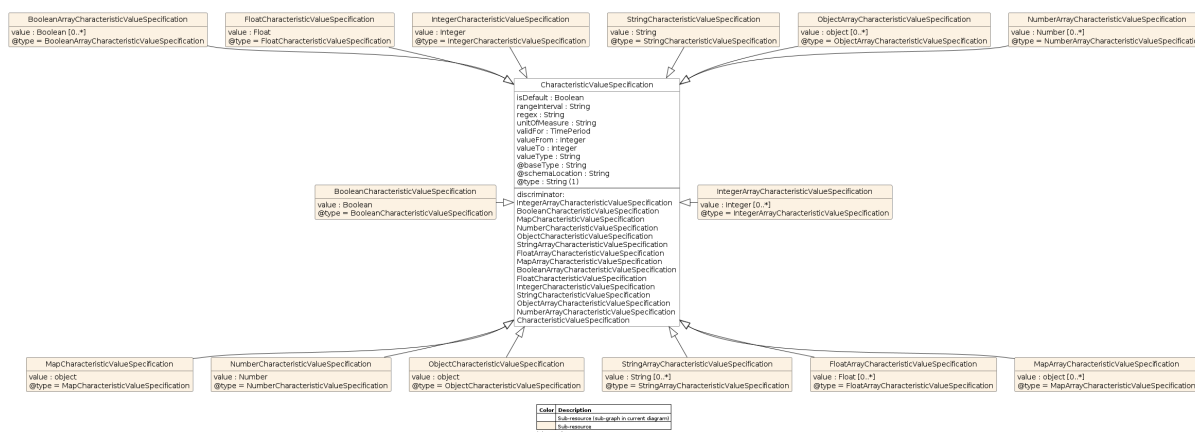


Figure 7 - CharacteristicValueSpecification

Field descriptions

PermissionSpecification fields

action	A String. Level of access granted as part of the permission, such as read or write.
description	A String. A fuller description of the permission specification, should describe to a human user what the purpose is of this permission.
function	A String. The function to which the action applies, could be an entity class name, in which case the action defines permission to read or modify these entities. E.g. Read ProductOffering. Alternatively, the function is an agreed string that reflects a specific functionality to which access is controlled, such as giving an ad-hoc price reduction in a product order.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. A descriptive name of the permission specification, should summarize to a human user what the purpose is of this permission.
specificationCharacteristic	A CharacteristicSpecification. List of characteristics that are relevant for evaluating the permission. For example, if the permission relates to granting an ad-hoc order price reduction, a characteristic might be defined for the maximum allowed reduction.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicSpecification sub-resource fields

See [\[CharacteristicSpecification\]](#) for details.

CharacteristicSpecificationRelationship sub-resource fields

See [\[CharacteristicSpecificationRelationship\]](#) for details.

CharacteristicValueSpecification sub-resource fields

See [\[CharacteristicValueSpecification\]](#) for details.

BooleanArrayCharacteristicValueSpecification sub-resource fields

See [\[BooleanArrayCharacteristicValueSpecification\]](#) for details.

BooleanCharacteristicValueSpecification sub-resource fields

See [\[BooleanCharacteristicValueSpecification\]](#) for details.

FloatArrayCharacteristicValueSpecification sub-resource fields

See [\[FloatArrayCharacteristicValueSpecification\]](#) for details.

FloatCharacteristicValueSpecification sub-resource fields

See [\[FloatCharacteristicValueSpecification\]](#) for details.

IntegerArrayCharacteristicValueSpecification sub-resource fields

See [\[IntegerArrayCharacteristicValueSpecification\]](#) for details.

IntegerCharacteristicValueSpecification sub-resource fields

See [\[IntegerCharacteristicValueSpecification\]](#) for details.

MapArrayCharacteristicValueSpecification sub-resource fields

See [\[MapArrayCharacteristicValueSpecification\]](#) for details.

MapCharacteristicValueSpecification sub-resource fields

See [\[MapCharacteristicValueSpecification\]](#) for details.

NumberArrayCharacteristicValueSpecification sub-resource fields

See [\[NumberArrayCharacteristicValueSpecification\]](#) for details.

NumberCharacteristicValueSpecification sub-resource fields

See [\[NumberCharacteristicValueSpecification\]](#) for details.

ObjectArrayCharacteristicValueSpecification sub-resource fields

See [\[ObjectArrayCharacteristicValueSpecification\]](#) for details.

ObjectCharacteristicValueSpecification sub-resource fields

See [\[ObjectCharacteristicValueSpecification\]](#) for details.

StringArrayCharacteristicValueSpecification sub-resource fields

See [\[StringArrayCharacteristicValueSpecification\]](#) for details.

StringCharacteristicValueSpecification sub-resource fields

See [\[StringCharacteristicValueSpecification\]](#) for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'PermissionSpecification' resource object.

```
{
  "@type": "PermissionSpecification",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-7e0a2d18",
  "id": "ps-7e0a2d18",
  "name": "ProductOrder:grantAlteration",
  "function": "ProductOrder",
  "action": "grantAlteration",
  "specificationCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "reductionPercentage",
      "valueType": "number",
      "characteristicValueSpecification": [
        {
          "@type": "FloatCharacteristicValueSpecification",
          "valueType": "number",
          "value": 5.5
        }
      ]
    }
  ]
}
```

CheckPermission resource

The CheckPermission task entity verifies that the security principal in context can carry out the action. If permission is granted, 200 is returned, if denied, 403 is returned. If party role or user is present in input, these are used, otherwise the security principal is deduced from the HTTP Authorization header.

Resource model

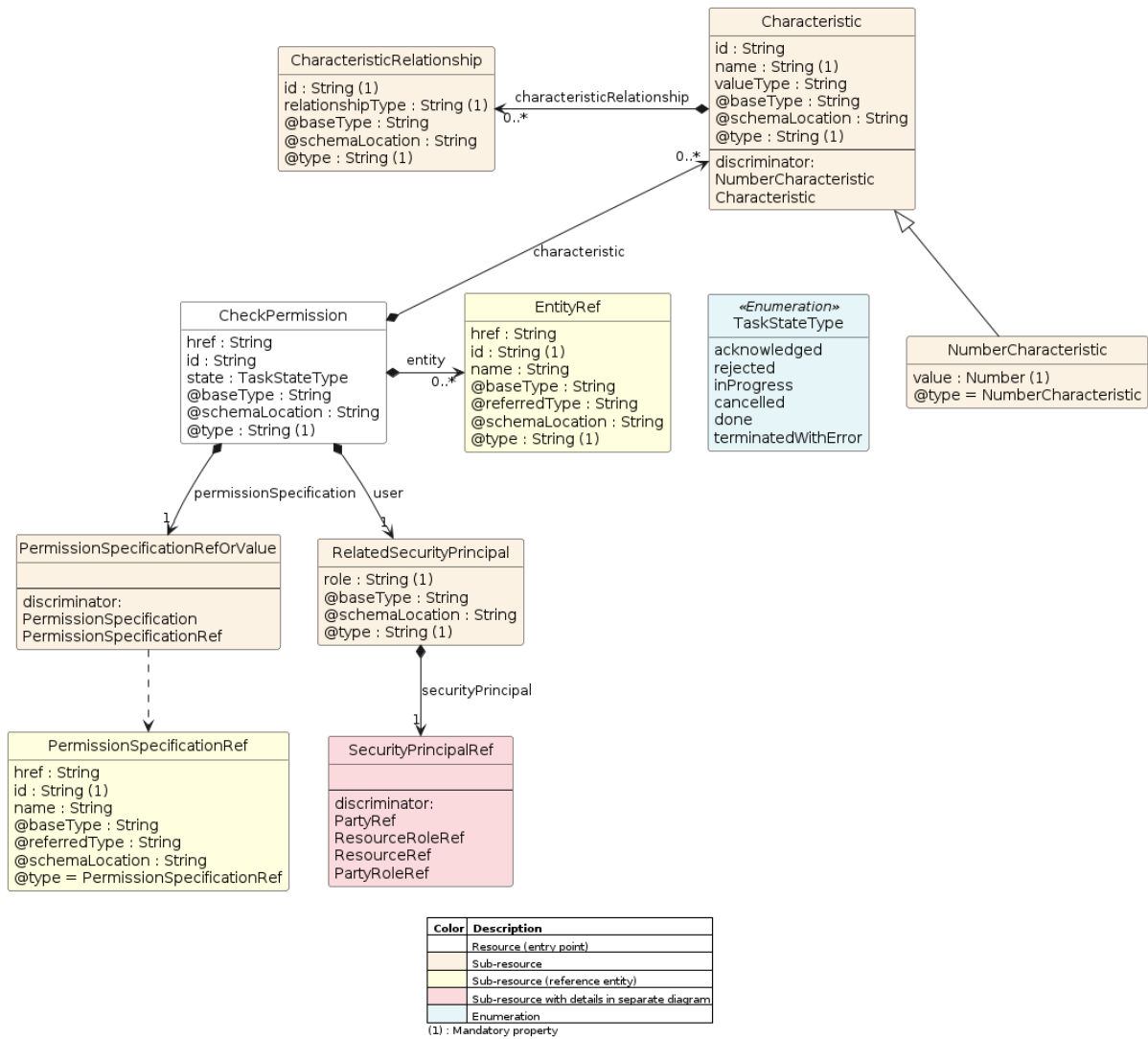


Figure 8 - CheckPermission

See [Figure 3](#) for SecurityPrincipalRef diagrams.

Field descriptions

CheckPermission fields

characteristic	A Characteristic. Input: Additional information that might be needed for a particular permission. For example, if Action is GiveDiscount on Function: ProductOrderItem, a characteristic could be used to transmit the percentage discount being asked for. Characteristic can be instantiated as * NumberCharacteristic
----------------	---

entity	An EntityRef. Input: A list of entities for which permission is being tested. For example, if Action is Update and Function is Customer, this could hold reference to specific customer to be updated. Or, if Action is Create and Function is ProductOffering, this could hold the entire entity, so that the permission could be checked against the line of business. Output: A list of entities for which permission is granted.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
permissionSpecification	A PermissionSpecificationRefOrValue. Related Entity reference. A related PermissionSpecification defines a PermissionSpecification described by reference or by value linked to a specific entity. The polymorphic attributes @type, @schemaLocation & @referredType are related to the PermissionSpecification entity and not the PermissionSpecificationRefOrValue class itself.
state	A TaskStateType. Possible values for the state of a task. ENUMERATED with values: * acknowledged * rejected * inProgress * cancelled * done * terminatedWithError
user	A RelatedSecurityPrincipal. A RelatedSecurityPrincipal defines a SecurityPrincipal reference, linked to a specific entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

See [\[Characteristic\]](#) for details.

CharacteristicRelationship sub-resource fields

See [\[CharacteristicRelationship\]](#) for details.

NumberCharacteristic sub-resource fields

See [\[NumberCharacteristic\]](#) for details.

EntityRef sub-resource fields

See [\[EntityRef\]](#) for details.

PermissionSpecification sub-resource fields

See [\[PermissionSpecification\]](#) for details.

CharacteristicSpecification sub-resource fields

See [\[CharacteristicSpecification\]](#) for details.

CharacteristicSpecificationRelationship sub-resource fields

See [\[CharacteristicSpecificationRelationship\]](#) for details.

CharacteristicValueSpecification sub-resource fields

See [\[CharacteristicValueSpecification\]](#) for details.

BooleanArrayCharacteristicValueSpecification sub-resource fields

See [\[BooleanArrayCharacteristicValueSpecification\]](#) for details.

BooleanCharacteristicValueSpecification sub-resource fields

See [\[BooleanCharacteristicValueSpecification\]](#) for details.

FloatArrayCharacteristicValueSpecification sub-resource fields

See [\[FloatArrayCharacteristicValueSpecification\]](#) for details.

FloatCharacteristicValueSpecification sub-resource fields

See [\[FloatCharacteristicValueSpecification\]](#) for details.

IntegerArrayCharacteristicValueSpecification sub-resource fields

See [\[IntegerArrayCharacteristicValueSpecification\]](#) for details.

IntegerCharacteristicValueSpecification sub-resource fields

See [\[IntegerCharacteristicValueSpecification\]](#) for details.

MapArrayCharacteristicValueSpecification sub-resource fields

See [\[MapArrayCharacteristicValueSpecification\]](#) for details.

MapCharacteristicValueSpecification sub-resource fields

See [\[MapCharacteristicValueSpecification\]](#) for details.

NumberArrayCharacteristicValueSpecification sub-resource fields

See [\[NumberArrayCharacteristicValueSpecification\]](#) for details.

NumberCharacteristicValueSpecification sub-resource fields

See [\[NumberCharacteristicValueSpecification\]](#) for details.

ObjectArrayCharacteristicValueSpecification sub-resource fields

See [\[ObjectArrayCharacteristicValueSpecification\]](#) for details.

ObjectCharacteristicValueSpecification sub-resource fields

See [\[ObjectCharacteristicValueSpecification\]](#) for details.

StringArrayCharacteristicValueSpecification sub-resource fields

See [\[StringArrayCharacteristicValueSpecification\]](#) for details.

StringCharacteristicValueSpecification sub-resource fields

See [\[StringCharacteristicValueSpecification\]](#) for details.

PermissionSpecificationRef sub-resource fields

See [\[PermissionSpecificationRef\]](#) for details.

RelatedSecurityPrincipal sub-resource fields

See [\[RelatedSecurityPrincipal\]](#) for details.

PartyRef sub-resource fields

See [\[PartyRef\]](#) for details.

PartyRoleRef sub-resource fields

See [\[PartyRoleRef\]](#) for details.

ResourceRef sub-resource fields

See [\[ResourceRef\]](#) for details.

ResourceRoleRef sub-resource fields

See [\[ResourceRoleRef\]](#) for details.

Json representation sample(s)

We provide below a JSON representation as example of the 'CheckPermission' resource object.

```
{
  "id": "32434646753180",
  "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/checkPermission/32434646753180"
  ,
  "@type": "CheckPermission",
  "state": "done",
  "permissionSpecification": {
    "@type": "PermissionSpecification",
    "function": "ProductOffering",
```

```
    "action": "create"
  },
  "user": {
    "role": "ProductManager",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "Jonathan.Goldberg",
      "@type": "PartyRoleRef",
      "@referredType": "PartyRole"
    }
  }
}
```

Notification Resource Models

11 notifications are defined for this API.

Notifications related to PermissionSet:

- Create Event
- Delete Event
- Attribute Value Change Event

Notifications related to PermissionSpecificationSet:

- Create Event
- Delete Event
- Attribute Value Change Event

Notifications related to PermissionSpecification:

- Create Event
- Delete Event
- Attribute Value Change Event

Notifications related to CheckPermission:

- Create Event
- State Change Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

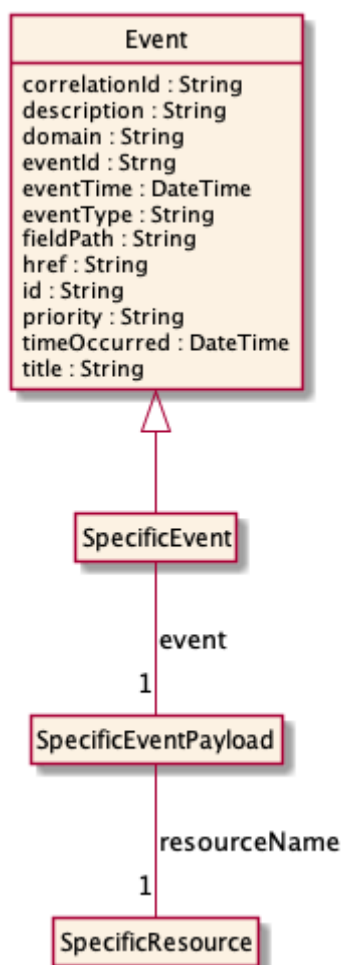


Figure 9 Notification Pattern

PermissionSet

Create Event

Message example for PermissionSetCreateEvent event

```

Content-Type: application/json
{
  "correlationId": "126-54f",
  "description": "PermissionSetCreateEvent illustration",
  "domain": "Infrastructure",
  "eventId": "125",
  "eventTime": "2021-09-27T07:43:59.059Z",
  "eventType": "PermissionSetCreateEvent",
  "priority": "1",
  "timeOccurred": "2021-09-27T07:43:59.059Z",
  "title": "PermissionSetCreateEvent",
  "event": {
    "permissionSet": {
      "@type": "PermissionSet",
      "href":
        "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSet/Prms-jnzgh456",
      "id": "Prms-jnzgh456",
      "creationDate": "2019-10-01T00:00:00.000Z",
    }
  }
}
  
```

```

    "description": "superProfile Granted for user",
    "validFor": {
      "startDateTime": "2019-10-01T00:00:00.000Z",
      "endDateTime": "2019-10-31T23:59:59.000Z"
    },
    "granter": {
      "role": "customer advisor",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "href": "https://serverRoot/partyManagement/v5/party/cclt-
456745",
        "id": "cclt-456745",
        "name": "Joan Smith",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    },
    "user": {
      "role": "client",
      "@type": "RelatedSecurityPrincipal",
      "securityPrincipal": {
        "id": "cust-745712-A",
        "href": "https://serverRoot/partyManagement/v5/party/cust-
745712-A",
        "name": "Jack Marshall",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    },
    "permission": [
      {
        "@type": "Permission",
        "permissionSpecification": {
          "@type": "PermissionSpecification",
          "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
31d5b770",
          "id": "ps-31d5b770",
          "name": "ProductOrder:update",
          "function": "ProductOrder",
          "action": "update"
        },
        "managedAssetGroup": {
          "@type": "FilterAssetGroup",
          "@baseType": "AssetGroup",
          "entityType": "ProductOrder",
          "filter":
"relatedParty.id=\cid3366116688\&relatedParty.@referredType=\Customer\"
        }
      }
    ]
  },
  "reportingSystem": {
    "id": "123",
    "name": "CRM app",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "123",
    "name": "CRM app",
    "@type": "ReportingResource",

```

```

    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "PermissionSetCreateEvent"
}

```

Delete Event

Message example for PermissionSetDeleteEvent event

```

Content-Type: application/json
{
  "correlationId": "145d-8hh",
  "description": "PermissionSetDeleteEvent illustration",
  "domain": "Commercial",
  "eventId": "523",
  "eventTime": "2021-09-27T07:43:59.059Z",
  "eventType": "PermissionSetDeleteEvent",
  "priority": "3",
  "timeOccurred": "2021-09-27T07:43:59.059Z",
  "title": "PermissionSetDeleteEvent",
  "event": {
    "permissionSet": {
      "@type": "PermissionSet",
      "href":
        "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSet/Prms-jnzgh456",
      "id": "Prms-jnzgh456"
    }
  },
  "reportingSystem": {
    "id": "123",
    "name": "IDM app",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "123",
    "name": "IDM app",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "PermissionSetDeleteEvent"
}

```

Attribute Value Change Event

Message example for AttributeValueChangeEvent event

```

Content-Type: application/json
{
  "correlationId": "333-fd6",
  "description": "PermissionSetAttributeValueChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "569",
  "eventTime": "2021-09-27T07:43:59.059Z",
  "eventType": "PermissionSetAttributeValueChangeEvent",

```

```

    "priority": "1",
    "timeOccurred": "2021-09-27T07:43:59.059Z",
    "title": "PermissionSetAttributeValueChangeEvent",
    "event": {
      "permissionSet": {
        "@type": "PermissionSet",
        "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSet/Prms-31d5b770",
        "id": "Prms-31d5b770",
        "validFor": {
          "startDateTime": "2020-10-01T00:00:00.000Z",
          "endDateTime": "2020-10-31T23:59:59.000Z"
        }
      }
    },
    "reportingSystem": {
      "id": "123",
      "name": "CRM app",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "123",
      "name": "CRM app",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "PermissionSetAttributeValueChangeEvent"
  }
}

```

PermissionSpecificationSet

Create Event

Message example for PermissionSpecificationSetCreateEvent event

```

Content-Type: application/json
{
  "correlationId": "126-54f",
  "description": "PermissionSpecificationSetCreateEvent illustration",
  "domain": "Commercial",
  "eventId": "125",
  "eventTime": "2021-09-27T07:43:59.059Z",
  "eventType": "PermissionSpecificationSetCreateEvent",
  "priority": "1",
  "timeOccurred": "2021-09-27T07:43:59.059Z",
  "title": "PermissionSpecificationSetCreateEvent",
  "event": {
    "permissionSpecificationSet": {
      "@type": "PermissionSpecificationSet",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-8f1a5628",
      "id": "pss-8f1a5628",
      "name": "Customer.OTT.Primary.User",
      "involvementRole": "Customer.OTT.Primary.User",
      "permissionSpecification": [
        {

```

```

        "@type": "PermissionSpecification",
        "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
a99c47c6",
        "id": "ps-a99c47c6",
        "description": "Permission to manage users and to add or remove
permissions to delegated users.",
        "name": "usermanagement:readwrite",
        "function": "user_management",
        "action": "ReadWrite"
    },
    {
        "@type": "PermissionSpecification",
        "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
5ea0f3cc",
        "id": "ps-5ea0f3cc",
        "description": "Permission to buy products in OTT apps.",
        "name": "impulsebuy:enabled",
        "function": "impulse_buy",
        "action": "Enabled"
    },
    {
        "@type": "PermissionSpecification",
        "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
b4b3601a",
        "id": "ps-b4b3601a",
        "description": "Permission to manage delegated users parental
controls.",
        "name": "parentalcontrols:manage",
        "function": "manage_parental_controls",
        "action": "ReadWrite"
    }
]
}
},
"reportingSystem": {
    "id": "123",
    "name": "CRM app",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
},
"source": {
    "id": "123",
    "name": "CRM app",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
},
"@baseType": "Event",
"@type": "PermissionSpecificationSetCreateEvent"
}

```

Delete Event

Message example for PermissionSpecificationSetDeleteEvent event

```

Content-Type: application/json
{
    "correlationId": "145d-8hh",

```

```

    "description": "PermissionSpecificationSetDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "523",
    "eventTime": "2021-09-27T07:43:59.059Z",
    "eventType": "PermissionSpecificationSetDeleteEvent",
    "priority": "3",
    "timeOccurred": "2021-09-27T07:43:59.059Z",
    "title": "PermissionSpecificationSetDeleteEvent",
    "event": {
      "permissionSpecificationSet": {
        "@type": "PermissionSpecificationSet",
        "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
ebe41def",
        "id": "pss-ebe41def"
      }
    },
    "reportingSystem": {
      "id": "123",
      "name": "CRM app",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "123",
      "name": "CRM app",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "PermissionSpecificationSetDeleteEvent"
  }

```

Attribute Value Change Event

Message example for AttributeValueChangeEvent event

```

Content-Type: application/json
{
  "correlationId": "akj-566",
  "description": "PermissionSpecificationSetAttributeValueChangeEvent
illustration",
  "domain": "Commercial",
  "eventId": "741",
  "eventTime": "2021-09-27T07:43:59.059Z",
  "eventType": "PermissionSpecificationSetAttributeValueChangeEvent",
  "priority": "1",
  "timeOccurred": "2021-09-27T07:43:59.059Z",
  "title": "PermissionSpecificationSetAttributeValueChangeEvent",
  "event": {
    "permissionSpecificationSet": {
      "@type": "PermissionSpecificationSet",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/4139
20",
      "id": "413920",
      "involvementRole": "MyTV SPORT+ familyAdmin Profile",
      "name": "TV Sport and Family Admin",
      "description": "Gives permissions to use Sport Channels and manage the
family"
    }
  }
}

```

```

    }
  },
  "reportingSystem": {
    "id": "123",
    "name": "CRM app",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "123",
    "name": "CRM app",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "PermissionSpecificationSetAttributeValueChangeEvent"
}

```

PermissionSpecification

Create Event

Message example for PermissionSpecificationCreateEvent event

```

Content-Type: application/json
{
  "correlationId": "126-54f",
  "description": "PermissionSpecificationCreateEvent illustration",
  "domain": "Commercial",
  "eventId": "125",
  "eventTime": "2021-09-27T07:43:59.059Z",
  "eventType": "PermissionSpecificationCreateEvent",
  "priority": "1",
  "timeOccurred": "2021-09-27T07:43:59.059Z",
  "title": "PermissionSpecificationCreateEvent",
  "event": {
    "permissionSpecification": {
      "@type": "PermissionSpecification",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-7e0a2d18",
      "id": "ps-7e0a2d18",
      "function": "ProductOrder",
      "action": "grantAlteration",
      "name": "Allow adhoc price alteration",
      "specificationCharacteristic": [
        {
          "@type": "CharacteristicSpecification",
          "name": "reductionPercentage",
          "valueType": "number",
          "characteristicValueSpecification": [
            {
              "@type": "FloatCharacteristicValueSpecification",
              "valueType": "number",
              "value": 5.5
            }
          ]
        }
      ]
    }
  }
}

```

```

    }
  },
  "reportingSystem": {
    "id": "123",
    "name": "CRM app",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "123",
    "name": "CRM app",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "PermissionSpecificationCreateEvent"
}

```

Delete Event

Message example for PermissionSpecificationDeleteEvent event

```

Content-Type: application/json
{
  "correlationId": "145d-8hh",
  "description": "PermissionSpecificationDeleteEvent illustration",
  "domain": "Commercial",
  "eventId": "523",
  "eventTime": "2021-09-27T07:43:59.059Z",
  "eventType": "PermissionSpecificationDeleteEvent",
  "priority": "3",
  "timeOccurred": "2021-09-27T07:43:59.059Z",
  "title": "PermissionSpecificationDeleteEvent",
  "event": {
    "permissionSpecification": {
      "@type": "PermissionSpecification",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-7e0a2d18",
      "id": "ps-7e0a2d18"
    }
  },
  "reportingSystem": {
    "id": "123",
    "name": "CRM app",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "123",
    "name": "CRM app",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "PermissionSpecificationDeleteEvent"
}

```

Attribute Value Change Event

Message example for AttributeValueChangeEvent event

```

Content-Type: application/json
{
  "correlationId": "akj-566",
  "description": "PermissionSpecificationAttributeValueChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "741",
  "eventTime": "2021-09-27T07:43:59.059Z",
  "eventType": "PermissionSpecificationAttributeValueChangeEvent",
  "priority": "1",
  "timeOccurred": "2021-09-27T07:43:59.059Z",
  "title": "PermissionSpecificationAttributeValueChangeEvent",
  "event": {
    "permissionSpecification": {
      "@type": "PermissionSpecification",
      "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-7e0a2d18",
      "id": "ps-7e0a2d18",
      "specificationCharacteristic": [
        {
          "@type": "CharacteristicSpecification",
          "name": "reductionPercentage",
          "valueType": "number",
          "characteristicValueSpecification": [
            {
              "@type": "FloatCharacteristicValueSpecification",
              "valueType": "number",
              "value": 5.5
            }
          ]
        }
      ]
    }
  },
  "reportingSystem": {
    "id": "123",
    "name": "CRM app",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "123",
    "name": "CRM app",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "PermissionSpecificationAttributeValueChangeEvent"
}

```

CheckPermission

Create Event

Message example for CheckPermissionCreateEvent event

```

Content-Type: application/json
{
  "correlationId": "0f874cb9-c70d",
  "description": "CheckPermissionCreateEvent illustration",
  "domain": "Commercial",
  "eventId": "41cf-ad81-d65ea1c2840c",
  "eventTime": "2022-08-25T12:18:12.191Z",
  "eventType": "CheckPermissionCreateEvent",
  "priority": "5",
  "timeOccurred": "2022-08-25T12:18:08.202Z",
  "title": "CheckPermissionCreateEvent",
  "event": {
    "checkPermission": {
      "id": "f3fc707ae9ce42fbbf273a46ccacdeca",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/checkPermission/f3fc707ae9ce42fbbf273a46ccacdeca",
      "@type": "CheckPermission",
      "state": "InProgress",
      "permissionSpecification": {
        "@type": "PermissionSpecification",
        "name": "ProductOffering::create",
        "function": "ProductOffering",
        "action": "create"
      },
      "user": {
        "role": "ProductManager",
        "@type": "RelatedSecurityPrincipal",
        "securityPrincipal": {
          "id": "Jonathan.Goldberg",
          "@type": "PartyRoleRef",
          "@referredType": "PartyRole"
        }
      }
    },
    "reportingSystem": {
      "id": "759",
      "name": "APP-745",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "705",
      "name": "APP-317",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "CheckPermissionCreateEvent"
  }
}

```

State Change Event

Message example for StateChangeEvent event

```
Content-Type: application/json
{
  "correlationId": "0f874cb9-c70d",
  "description": "CheckPermissionStateChangeEvent illustration",
  "domain": "Commercial",
  "eventId": "41cf-ad81-d65ea1c2840c",
  "eventTime": "2022-08-25T12:18:12.191Z",
  "eventType": "CheckPermissionStateChangeEvent",
  "priority": "5",
  "timeOccurred": "2022-08-25T12:18:08.202Z",
  "title": "CheckPermissionStateChangeEvent",
  "event": {
    "checkPermission": {
      "id": "32434646753180",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/checkPermission/32434646753180"
    },
    "@type": "CheckPermission",
    "state": "done"
  },
  "reportingSystem": {
    "id": "759",
    "name": "APP-745",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "source": {
    "id": "705",
    "name": "APP-317",
    "@type": "ReportingResource",
    "@referredType": "LogicalResource"
  },
  "@baseType": "Event",
  "@type": "CheckPermissionStateChangeEvent"
}
```

API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource.
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource.
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource.
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources.
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on PermissionSet

Retrieves a PermissionSet by ID

GET /permissionSet/{id}?fields=...

Description

This operation retrieves a PermissionSet entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a permission set by its identifier, showing how permission on order is limited to orders owned by a specific customer, using JSONPath filter

Request

```
GET /permissionSet/Prms-jnzgh456
Content-Type: application/json
```

Response

```
200
{
  "@type": "PermissionSet",
  "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSet/Prms-
```

```

jnzgh456",
  "id": "Prms-jnzgh456",
  "creationDate": "2019-10-01T00:00:00.000Z",
  "description": "superProfile Granted for user",
  "validFor": {
    "startDateTime": "2019-10-01T00:00:00.000Z",
    "endDateTime": "2019-10-31T23:59:59.000Z"
  },
  "user": {
    "role": "client",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "cust-745712-A",
      "href": "https://host:port/partyManagement/v5/party/cust-745712-A",
      "name": "Jack Marshall",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "granter": {
    "role": "customer advisor",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "cclt-456745",
      "href": "https://host:port/partyManagement/v5/party/cclt-456745",
      "name": "John Baker",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "permission": [
    {
      "@type": "Permission",
      "permissionSpecification": {
        "@type": "PermissionSpecification",
        "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-31d5b770",
        "id": "ps-31d5b770",
        "name": "ProductOrder:update",
        "function": "ProductOrder",
        "action": "update"
      },
      "managedAssetGroup": {
        "@baseType": "AssetGroup",
        "@type": "JSONPathAssetGroup",
        "entityType": "ProductOrder",
        "jsonPath": "filter=productSpecification=='tv-channel'&&productCharacteristic[?(@.name=='tvrating' && @.value=='children')]"
      }
    }
  ]
}

```

Retrieve a permission set by its identifier, showing simple permission for access to IPTV for the user

Request

```
GET /permissionSet/Prms-31d5b770
```

```
Content-Type: application/json
```

Response

```
200
{
  "@type": "PermissionSet",
  "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSet/Prms-31d5b770",
  "id": "Prms-31d5b770",
  "creationDate": "2019-10-01T00:00:00.000Z",
  "validFor": {
    "startDateTime": "2019-10-01T00:00:00.000Z",
    "endDateTime": "2019-10-31T23:59:59.000Z"
  },
  "user": {
    "role": "client",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "cust-745712-A",
      "href": "https://host:port/partyManagement/v5/party/cust-745712-A",
      "name": "Jack Marshall",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "permission": [
    {
      "@type": "Permission",
      "permissionSpecification": {
        "@type": "PermissionSpecification",
        "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-b2e2fa58",
        "id": "ps-b2e2fa58",
        "name": "IPTVaccess:enabled",
        "function": "IPTV access",
        "action": "enabled"
      }
    }
  ]
}
```

Retrieve a permission set by its identifier, showing how permissions are applied to a specific list of entities

Request

```
GET /permissionSet/63116b98
Content-Type: application/json
```

Response

```
200
{
```

```

"@type": "PermissionSet",
"href": "https://serverRoot/rolesAndPermissionsManagement/v5/permission/Prms-63116b98",
"id": "63116b98",
"creationDate": "2023-03-01T16:49:00.000Z",
"description": "Owner profile granted to user automatically for its owned customer, product and service entities.",
"validFor": {
  "startTime": "2023-03-01T00:00:00.000Z",
  "endTime": "2023-04-01T00:00:00.000Z"
},
"user": {
  "role": "customer advisor",
  "@type": "RelatedSecurityPrincipal",
  "securityPrincipal": {
    "id": "cclt-456745",
    "href": "https://host:port/partyManagement/v5/party/cclt-456745",
    "name": "John Baker",
    "@type": "PartyRef",
    "@referredType": "Individual"
  }
},
"permission": [
  {
    "@type": "Permission",
    "managedAssetGroup": {
      "@type": "ListAssetGroup",
      "@baseType": "AssetGroup",
      "entity": [
        {
          "@type": "EntityRef",
          "href": "https://serverRoot/customerManagement/v5/customer/C123456789",
          "id": "C123456789",
          "@referredType": "Customer"
        },
        {
          "@type": "EntityRef",
          "href": "https://serverRoot/customerManagement/v5/customer/5.99999.99.99.999999",
          "id": "5.99999.99.99.999999",
          "@referredType": "Customer"
        },
        {
          "@type": "EntityRef",
          "href": "https://serverRoot/serviceInventory/v5/service/S123456789",
          "id": "S123456789",
          "@referredType": "Service"
        },
        {
          "@type": "EntityRef",
          "href": "https://serverRoot/productInventory/v5/service/1-44up07",
          "id": "1-44up07",
          "@referredType": "Product"
        }
      ]
    }
  },
  {
    "@type": "PermissionSpecificationSet",
    "@type": "PermissionSpecificationSetRef",
    "@referredType": "PermissionSpecificationSet",
    "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-5de89f4f",
    "id": "pss-5de89f4f"
  }
]

```

```

    }
  }
}

```

Retrieve a permission set by its identifier, showing multiple permissions with different types of asset specifications

Request

```

GET /permissionSet/7a0cb197
Content-Type: application/json

```

Response

```

200
{
  "@type": "PermissionSet",
  "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permission/Prms-7a0cb197",
  "id": "7a0cb197",
  "creationDate": "2023-03-01T16:49:00.000Z",
  "description": "Customer selfcare admin profile granted by owner to delegate management of customers, products and services entities.",
  "validFor": {
    "startDateTime": "2023-03-01T00:00:00.000Z",
    "endDateTime": "2023-04-01T00:00:00.000Z"
  },
  "user": {
    "role": "client",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "cust-745712-A",
      "href": "https://host:port/partyManagement/v5/party/cust-745712-A",
      "name": "Jack Marshall",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "granter": {
    "role": "customer advisor",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "p7654321",
      "href": "https://host:port/partyManagement/v5/party/p7654321",
      "name": "Lynda Owens",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "permission": [
    {
      "@type": "Permission",
      "managedAssetGroup": {
        "@type": "JSONPathAssetGroup",
        "@baseType": "AssetGroup",
        "entityType": "Customer",

```

```

    "jsonPath": "$[status=='Active']"
  },
  "permissionSpecificationSet": {
    "@type": "PermissionSpecificationSetRef",
    "@referredType": "PermissionSpecificationSet",
    "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
edd9d8a4",
    "id": "pss-edd9d8a4"
  }
},
{
  "@type": "Permission",
  "managedAssetGroup": {
    "@type": "ListAssetGroup",
    "@baseType": "AssetGroup",
    "entity": [
      {
        "@type": "EntityRef",
        "href":
"https://serverRoot/customerManagement/v5/customer/5.99999.99.99.999999",
        "id": "5.99999.99.99.999999",
        "@referredType": "Customer"
      }
    ]
  },
  "permissionSpecificationSet": {
    "@type": "PermissionSpecificationSetRef",
    "@referredType": "PermissionSpecificationSet",
    "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
edd9d8a4",
    "id": "pss-edd9d8a4"
  }
},
{
  "@type": "Permission",
  "managedAssetGroup": {
    "@type": "ListAssetGroup",
    "@baseType": "AssetGroup",
    "entity": [
      {
        "@type": "EntityRef",
        "href": "https://serverRoot/productInventory/v5/service/1-44up07",
        "id": "1-44up07",
        "@referredType": "Product"
      }
    ]
  },
  "permissionSpecificationSet": {
    "@type": "PermissionSpecificationSetRef",
    "@referredType": "PermissionSpecificationSet",
    "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
edd9d8a4",
    "id": "pss-edd9d8a4"
  }
},
{
  "@type": "Permission",
  "managedAssetGroup": {
    "@type": "ListAssetGroup",
    "@baseType": "AssetGroup",

```

```

    "entity": [
      {
        "@type": "EntityRef",
        "href": "https://serverRoot/serviceInventory/v5/service/S123456789",
        "id": "S123456789",
        "@referredType": "Service"
      }
    ],
    "permissionSpecificationSet": {
      "@type": "PermissionSpecificationSetRef",
      "@referredType": "PermissionSpecificationSet",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
edd9d8a4",
      "id": "pss-edd9d8a4"
    }
  }
]
}

```

Retrieve a permission set by its identifier, showing multiple permissions with different types of asset specifications

Request

```

GET /permissionSet/7a0cb197-381a-4043-9cd6-aaef3729589f
Content-Type: application/json

```

Response

```

200
{
  "@type": "PermissionSet",
  "href": "https://server:port/userRolePermission/v5/permissionSet/7a0cb197-381a-
4043-9cd6-aaef3729589f",
  "id": "7a0cb197-381a-4043-9cd6-aaef3729589f",
  "creationDate": "2023-03-01T16:49:00.000Z",
  "description": "Customer OTT primary user profile granted by owner",
  "validFor": {
    "startDateTime": "2023-03-01T00:00:00.000Z",
    "endDateTime": "2023-04-01T00:00:00.000Z"
  },
  "user": {
    "role": "client",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "cust-745712-A",
      "href": "https://host:port/partyManagement/v5/party/cust-745712-A",
      "name": "Jack Marshall",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "granter": {
    "role": "customer advisor",
    "@type": "RelatedPartyRefOrPartyRoleRef",

```

```

    "partyOrPartyRole": {
      "id": "cclt-456745",
      "href": "https://host:port/partyManagement/v5/party/cclt-456745",
      "name": "John Baker",
      "@type": "PartyRef",
      "@referredType": "Individual"
    },
    "permission": [
      {
        "@type": "Permission",
        "managedAssetGroup": {
          "@type": "ListAssetGroup",
          "@baseType": "AssetGroup",
          "entity": [
            {
              "id": "S123456789",
              "@type": "EntityRef",
              "@referredType": "ServiceAccount"
            }
          ]
        },
        "permissionSpecificationSet": {
          "@type": "PermissionSpecificationSetRef",
          "href":
            "https://server:port/userRolePermission/v5/permissionSpecificationSet/adf0487a-1671-49c1-9be1-87e66a6d3a69",
          "id": "adf0487a-1671-49c1-9be1-87e66a6d3a69",
          "@referredType": "PermissionSpecificationSet"
        }
      }
    ]
  }
}

```

Retrieve a permission set by its identifier, showing multiple permissions with different types of asset specifications

Request

```

GET /permissionSet/7a0cb197-381a-4043-9cd6-aaef3729589f
Content-Type: application/json

```

Response

```

200
{
  "@type": "PermissionSet",
  "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSet/pst-9c512f49",
  "id": "pst-9c512f49",
  "creationDate": "2025-10-01T00:00:00.000Z",
  "description": "Product Catalog Management webapp TMF620 API consumer role permission set",
  "validFor": {
    "startDateTime": "2025-10-01T00:00:00.000Z",
    "endDateTime": "2025-10-31T23:59:59.000Z"
  },
}

```

```

"user": {
  "role": "tmf-api-consumer",
  "@type": "RelatedSecurityPrincipal",
  "securityPrincipal": {
    "id": "rr-82288122",
    "href": "https://host:port/resourceRoleManagement/v5/resourceRole/rr-82288122",
    "name": "Product Catalog Management Web Application TMF Open API Consumer Role",
    "@type": "ResourceRoleRef",
    "@referredType": "ResourceRole"
  }
},
"granter": {
  "role": "product_catalog_mng_admin",
  "@type": "RelatedPartyRefOrPartyRoleRef",
  "partyOrPartyRole": {
    "id": "p-70fabcac9b01",
    "href": "https://host:port/partyManagement/v5/party/p-70fabcac9b01",
    "name": "Darth Vader",
    "@type": "PartyRef",
    "@referredType": "Individual"
  }
},
"permission": [
  {
    "@type": "Permission",
    "permissionSpecification": {
      "@type": "PermissionSpecification",
      "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-31d5b770",
      "id": "psp-31d5b770",
      "name": "Catalog:ReadWrite",
      "function": "Catalog",
      "action": "ReadWrite"
    },
    "managedAssetGroup": {
      "@type": "SetAssetGroup",
      "@baseType": "AssetGroup",
      "name": "Catalog Asset Group Set",
      "entityType": [
        "Catalog"
      ]
    }
  },
  {
    "@type": "Permission",
    "permissionSpecification": {
      "@type": "PermissionSpecification",
      "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-d9f344d5",
      "id": "psp-d9f344d5",
      "name": "ProductOffering:ReadWrite",
      "function": "ProductOffering",
      "action": "ReadWrite"
    },
    "managedAssetGroup": {
      "@type": "SetAssetGroup",
      "@baseType": "AssetGroup",
      "name": "ProductOffering Asset Group Set",
      "entityType": [

```

```

        "ProductOffering"
      ]
    }
  },
  {
    "@type": "Permission",
    "permissionSpecification": {
      "@type": "PermissionSpecification",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-7fa60a0f",
      "id": "psp-7fa60a0f",
      "name": "ProductOfferingPrice:ReadWrite",
      "function": "ProductOfferingPrice",
      "action": "ReadWrite"
    },
    "managedAssetGroup": {
      "@type": "SetAssetGroup",
      "@baseType": "AssetGroup",
      "name": "ProductOfferingPrice Asset Group Set",
      "entityType": [
        "ProductOfferingPrice"
      ]
    }
  }
]
}

```

List or find PermissionSet objects

GET /permissionSet?fields=...

Description

This operation list PermissionSet entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a list of permission sets

Request

```

GET /permissionSet
Content-Type: application/json

```

Response

```

200
[
  {
    "@type": "PermissionSet",
    "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSet/Prms-jnzgh456",

```

```

    "id": "Prms-jnzgh456",
    "creationDate": "2019-10-01T00:00:00.000Z",
    "description": "superProfile Granted for user",
    "validFor": {
      "startDateTime": "2019-10-01T00:00:00.000Z",
      "endDateTime": "2019-10-31T23:59:59.000Z"
    },
    "user": {
      "role": "client",
      "@type": "RelatedSecurityPrincipal",
      "securityPrincipal": {
        "id": "cust-745712-A",
        "href": "https://host:port/partyManagement/v5/party/cust-745712-A",
        "name": "Jack Marshall",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    },
    "granter": {
      "role": "customer advisor",
      "@type": "RelatedPartyRefOrPartyRoleRef",
      "partyOrPartyRole": {
        "id": "cclt-456745",
        "href": "https://host:port/partyManagement/v5/party/cclt-456745",
        "name": "John Baker",
        "@type": "PartyRef",
        "@referredType": "Individual"
      }
    },
    "permission": [
      {
        "@type": "Permission",
        "permissionSpecification": {
          "@type": "PermissionSpecification",
          "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-31d5b770",
          "id": "ps-31d5b770",
          "name": "ProductOrder:update",
          "function": "ProductOrder",
          "action": "update"
        },
        "managedAssetGroup": {
          "@baseType": "AssetGroup",
          "@type": "JSONPathAssetGroup",
          "entityType": "ProductOrder",
          "jsonPath": "filter=productSpecification=='tv-channel'&productCharacteristic[?(@.name=='tvrating' && @.value=='children')]"
        }
      }
    ],
    {
      "@type": "PermissionSet",
      "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSet/Prms-31d5b770",
      "id": "Prms-31d5b770",
      "creationDate": "2019-10-01T00:00:00.000Z",
      "validFor": {
        "startDateTime": "2019-10-01T00:00:00.000Z",
        "endDateTime": "2019-10-31T23:59:59.000Z"
      },
      "user": {

```

```

    "role": "client",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "cust-745712-A",
      "href": "https://host:port/partyManagement/v5/party/cust-745712-A",
      "name": "Jack Marshall",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "permission": [
    {
      "@type": "Permission",
      "permissionSpecification": {
        "@type": "PermissionSpecification",
        "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
b2e2fa58",
        "id": "ps-b2e2fa58",
        "name": "IPTVaccess:enabled",
        "function": "IPTV access",
        "action": "enabled"
      }
    }
  ]
},
{
  "@type": "PermissionSet",
  "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permission/Prms-
63116b98",
  "id": "63116b98",
  "creationDate": "2023-03-01T16:49:00.000Z",
  "description": "Owner profile granted to user automatically for its owned
customer, product and service entities.",
  "validFor": {
    "startDateTime": "2023-03-01T00:00:00.000Z",
    "endDateTime": "2023-04-01T00:00:00.000Z"
  },
  "user": {
    "role": "customer advisor",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "cclt-456745",
      "href": "https://host:port/partyManagement/v5/party/cclt-456745",
      "name": "John Baker",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "permission": [
    {
      "@type": "Permission",
      "managedAssetGroup": {
        "@type": "ListAssetGroup",
        "@baseType": "AssetGroup",
        "entity": [
          {
            "@type": "EntityRef",
            "href":
"https://serverRoot/customerManagement/v5/customer/C123456789",
            "id": "C123456789",
            "@referredType": "Customer"
          }
        ]
      }
    }
  ]
},

```

```

        {
            "@type": "EntityRef",
            "href":
"https://serverRoot/customerManagement/v5/customer/5.99999.99.99.999999",
            "id": "5.99999.99.99.999999",
            "@referredType": "Customer"
        },
        {
            "@type": "EntityRef",
            "href": "https://serverRoot/serviceInventory/v5/service/S123456789",
            "id": "S123456789",
            "@referredType": "Service"
        },
        {
            "@type": "EntityRef",
            "href": "https://serverRoot/productInventory/v5/service/1-44up07",
            "id": "1-44up07",
            "@referredType": "Product"
        }
    ]
},
"permissionSpecificationSet": {
    "@type": "PermissionSpecificationSetRef",
    "@referredType": "PermissionSpecificationSet",
    "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
5de89f4f",
    "id": "pss-5de89f4f"
}
}
],
{
    "@type": "PermissionSet",
    "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permission/Prms-
7a0cb197",
    "id": "7a0cb197",
    "creationDate": "2023-03-01T16:49:00.000Z",
    "description": "Customer selfcare admin profile granted by owner to delegate
management of customers, products and services entities.",
    "validFor": {
        "startDateTime": "2023-03-01T00:00:00.000Z",
        "endDateTime": "2023-04-01T00:00:00.000Z"
    },
    "user": {
        "role": "client",
        "@type": "RelatedSecurityPrincipal",
        "securityPrincipal": {
            "id": "cust-745712-A",
            "href": "https://host:port/partyManagement/v5/party/cust-745712-A",
            "name": "Jack Marshall",
            "@type": "PartyRef",
            "@referredType": "Individual"
        }
    },
    "granter": {
        "role": "customer advisor",
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "partyOrPartyRole": {
            "id": "p7654321",
            "href": "https://host:port/partyManagement/v5/party/p7654321",
            "name": "Lynda Owens",
            "@type": "PartyRef",

```

```

    "@referredType": "Individual"
  }
},
"permission": [
  {
    "@type": "Permission",
    "managedAssetGroup": {
      "@type": "JSONPathAssetGroup",
      "@baseType": "AssetGroup",
      "entityType": "Customer",
      "jsonPath": "$[status=='Active']"
    },
    "permissionSpecificationSet": {
      "@type": "PermissionSpecificationSetRef",
      "@referredType": "PermissionSpecificationSet",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
edd9d8a4",
      "id": "pss-edd9d8a4"
    }
  },
  {
    "@type": "Permission",
    "managedAssetGroup": {
      "@type": "ListAssetGroup",
      "@baseType": "AssetGroup",
      "entity": [
        {
          "@type": "EntityRef",
          "href":
"https://serverRoot/customerManagement/v5/customer/5.99999.99.99.999999",
          "id": "5.99999.99.99.999999",
          "@referredType": "Customer"
        }
      ]
    },
    "permissionSpecificationSet": {
      "@type": "PermissionSpecificationSetRef",
      "@referredType": "PermissionSpecificationSet",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
edd9d8a4",
      "id": "pss-edd9d8a4"
    }
  },
  {
    "@type": "Permission",
    "managedAssetGroup": {
      "@type": "ListAssetGroup",
      "@baseType": "AssetGroup",
      "entity": [
        {
          "@type": "EntityRef",
          "href": "https://serverRoot/productInventory/v5/service/1-44up07",
          "id": "1-44up07",
          "@referredType": "Product"
        }
      ]
    },
    "permissionSpecificationSet": {
      "@type": "PermissionSpecificationSetRef",
      "@referredType": "PermissionSpecificationSet",
      "href":

```

```

"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
edd9d8a4",
  "id": "pss-edd9d8a4"
}
},
{
  "@type": "Permission",
  "managedAssetGroup": {
    "@type": "ListAssetGroup",
    "@baseType": "AssetGroup",
    "entity": [
      {
        "@type": "EntityRef",
        "href": "https://serverRoot/serviceInventory/v5/service/S123456789",
        "id": "S123456789",
        "@referredType": "Service"
      }
    ]
  },
  "permissionSpecificationSet": {
    "@type": "PermissionSpecificationSetRef",
    "@referredType": "PermissionSpecificationSet",
    "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
edd9d8a4",
    "id": "pss-edd9d8a4"
  }
}
],
},
{
  "@type": "PermissionSet",
  "href": "https://server:port/userRolePermission/v5/permissionSet/7a0cb197-381a-
4043-9cd6-aaef3729589f",
  "id": "7a0cb197-381a-4043-9cd6-aaef3729589f",
  "creationDate": "2023-03-01T16:49:00.000Z",
  "description": "Customer OTT primary user profile granted by owner",
  "validFor": {
    "startDateTime": "2023-03-01T00:00:00.000Z",
    "endDateTime": "2023-04-01T00:00:00.000Z"
  },
  "user": {
    "role": "client",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "cust-745712-A",
      "href": "https://host:port/partyManagement/v5/party/cust-745712-A",
      "name": "Jack Marshall",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "granter": {
    "role": "customer advisor",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "cclt-456745",
      "href": "https://host:port/partyManagement/v5/party/cclt-456745",
      "name": "John Baker",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  }
},
},

```

```

    "permission": [
      {
        "@type": "Permission",
        "managedAssetGroup": {
          "@type": "ListAssetGroup",
          "@baseType": "AssetGroup",
          "entity": [
            {
              "id": "S123456789",
              "@type": "EntityRef",
              "@referredType": "ServiceAccount"
            }
          ]
        },
        "permissionSpecificationSet": {
          "@type": "PermissionSpecificationSetRef",
          "href": "https://server:port/userRolePermission/v5/permissionSpecificationSet/adf0487a-1671-49c1-9be1-87e66a6d3a69",
          "id": "adf0487a-1671-49c1-9be1-87e66a6d3a69",
          "@referredType": "PermissionSpecificationSet"
        }
      }
    ],
    {
      "@type": "PermissionSet",
      "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSet/pst-9c512f49",
      "id": "pst-9c512f49",
      "creationDate": "2025-10-01T00:00:00.000Z",
      "description": "Product Catalog Management webapp TMF620 API consumer role permission set",
      "validFor": {
        "startDateTime": "2025-10-01T00:00:00.000Z",
        "endDateTime": "2025-10-31T23:59:59.000Z"
      },
      "user": {
        "role": "tmf-api-consumer",
        "@type": "RelatedSecurityPrincipal",
        "securityPrincipal": {
          "id": "rr-82288122",
          "href": "https://host:port/resourceRoleManagement/v5/resourceRole/rr-82288122",
          "name": "Product Catalog Management Web Application TMF Open API Consumer Role",
          "@type": "ResourceRoleRef",
          "@referredType": "ResourceRole"
        }
      },
      "granter": {
        "role": "product_catalog_mng_admin",
        "@type": "RelatedPartyRefOrPartyRoleRef",
        "partyOrPartyRole": {
          "id": "p-70fabcac9b01",
          "href": "https://host:port/partyManagement/v5/party/p-70fabcac9b01",
          "name": "Darth Vader",
          "@type": "PartyRef",
          "@referredType": "Individual"
        }
      }
    },
    "permission": [
      {

```

```

    "@type": "Permission",
    "permissionSpecification": {
      "@type": "PermissionSpecification",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-31d5b770",
      "id": "psp-31d5b770",
      "name": "Catalog:ReadWrite",
      "function": "Catalog",
      "action": "ReadWrite"
    },
    "managedAssetGroup": {
      "@type": "SetAssetGroup",
      "@baseType": "AssetGroup",
      "name": "Catalog Asset Group Set",
      "entityType": [
        "Catalog"
      ]
    }
  },
  {
    "@type": "Permission",
    "permissionSpecification": {
      "@type": "PermissionSpecification",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-d9f344d5",
      "id": "psp-d9f344d5",
      "name": "ProductOffering:ReadWrite",
      "function": "ProductOffering",
      "action": "ReadWrite"
    },
    "managedAssetGroup": {
      "@type": "SetAssetGroup",
      "@baseType": "AssetGroup",
      "name": "ProductOffering Asset Group Set",
      "entityType": [
        "ProductOffering"
      ]
    }
  },
  {
    "@type": "Permission",
    "permissionSpecification": {
      "@type": "PermissionSpecification",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-7fa60a0f",
      "id": "psp-7fa60a0f",
      "name": "ProductOfferingPrice:ReadWrite",
      "function": "ProductOfferingPrice",
      "action": "ReadWrite"
    },
    "managedAssetGroup": {
      "@type": "SetAssetGroup",
      "@baseType": "AssetGroup",
      "name": "ProductOfferingPrice Asset Group Set",
      "entityType": [
        "ProductOfferingPrice"
      ]
    }
  }
]

```

```
}
]
```

Creates a PermissionSet

POST /permissionSet?fields=...

Description

This operation creates a PermissionSet entity.

Mandatory Attributes

Mandatory Attributes	Rule
permission	
user	
@type	

Usage samples

Creation of a new PermissionSet with POST operation

Request

```
POST /permissionSet
Content-Type: application/json
{
  "@type": "PermissionSet",
  "description": "superProfile Granted for user",
  "validFor": {
    "startDateTime": "2019-10-01T00:00:00.000Z",
    "endDateTime": "2019-10-31T23:59:59.000Z"
  },
  "granter": {
    "role": "customer advisor",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "cclt-456745",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "user": {
    "role": "client",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "cust-745712-A",
      "name": "Jack Marshall",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "permission": [
    {
      "@type": "Permission",
```

```

    "permissionSpecification": {
      "@type": "PermissionSpecification",
      "name": "ProductOrder:update",
      "function": "ProductOrder",
      "action": "update"
    },
    "managedAssetGroup": {
      "@type": "FilterAssetGroup",
      "@baseType": "AssetGroup",
      "entityType": "ProductOrder",
      "filter":
"relatedParty.id=\"cid3366116688\"&relatedParty.@referredType=\"Customer\""
    }
  }
]
}

```

Response

```

201
{
  "@type": "PermissionSet",
  "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSet/Prms-jnzgh456",
  "id": "Prms-jnzgh456",
  "creationDate": "2019-10-01T00:00:00.000Z",
  "description": "superProfile Granted for user",
  "validFor": {
    "startDateTime": "2019-10-01T00:00:00.000Z",
    "endDateTime": "2019-10-31T23:59:59.000Z"
  },
  "granter": {
    "role": "customer advisor",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "href": "https://serverRoot/partyManagement/v5/party/cclt-456745",
      "id": "cclt-456745",
      "name": "Joan Smith",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "user": {
    "role": "client",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "cust-745712-A",
      "href": "https://serverRoot/partyManagement/v5/party/cust-745712-A",
      "name": "Jack Marshall",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "permission": [
    {
      "@type": "Permission",
      "permissionSpecification": {
        "@type": "PermissionSpecification",
        "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-

```

```

31d5b770",
  "id": "ps-31d5b770",
  "name": "ProductOrder:update",
  "function": "ProductOrder",
  "action": "update"
},
"managedAssetGroup": {
  "@type": "FilterAssetGroup",
  "@baseType": "AssetGroup",
  "entityType": "ProductOrder",
  "filter":
"relatedParty.id=\"cid3366116688\"&relatedParty.@referredType=\"Customer\""
}
}
]
}

```

Creation of a new PermissionSet with POST operation

Request

```

POST /permissionSet
Content-Type: application/json
{
  "@type": "PermissionSet",
  "validFor": {
    "startDateTime": "2019-10-01T00:00:00.000Z",
    "endDateTime": "2019-10-31T23:59:59.000Z"
  },
  "user": {
    "role": "client",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "cust-745712-A",
      "href": "https://host:port/partyManagement/v5/party/cust-745712-A",
      "name": "Jack Marshall",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "permission": [
    {
      "@type": "Permission",
      "permissionSpecification": {
        "@type": "PermissionSpecification",
        "name": "IPITVaccess:enabled",
        "function": "IPTV access",
        "action": "enabled"
      }
    }
  ]
}

```

Response

```

201
{

```

```

"@type": "PermissionSet",
"href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSet/Prms-31d5b770",
"id": "Prms-31d5b770",
"creationDate": "2019-10-01T00:00:00.000Z",
"validFor": {
  "startDateTime": "2019-10-01T00:00:00.000Z",
  "endDateTime": "2019-10-31T23:59:59.000Z"
},
"user": {
  "role": "client",
  "@type": "RelatedSecurityPrincipal",
  "securityPrincipal": {
    "id": "cust-745712-A",
    "href": "https://host:port/partyManagement/v5/party/cust-745712-A",
    "name": "Jack Marshall",
    "@type": "PartyRef",
    "@referredType": "Individual"
  }
},
"permission": [
  {
    "@type": "Permission",
    "permissionSpecification": {
      "@type": "PermissionSpecification",
      "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-b2e2fa58",
      "id": "ps-b2e2fa58",
      "name": "IPITVaccess:enabled",
      "function": "IPTV access",
      "action": "enabled"
    }
  }
]
}

```

Creation of a new PermissionSet with POST operation

Request

```

POST /permissionSet
Content-Type: application/json
{
  "@type": "PermissionSet",
  "description": "Owner profile granted to user automatically for its owned customer, product and service entities.",
  "validFor": {
    "startDateTime": "2023-03-01T00:00:00.000Z",
    "endDateTime": "2023-04-01T00:00:00.000Z"
  },
  "user": {
    "role": "family member",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "cclt-456745",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  }
},

```

```

"permission": [
  {
    "@type": "Permission",
    "managedAssetGroup": {
      "@type": "ListAssetGroup",
      "@baseType": "AssetGroup",
      "entity": [
        {
          "@type": "EntityRef",
          "id": "C123456789",
          "@referredType": "Customer"
        },
        {
          "@type": "EntityRef",
          "id": "5.99999.99.99.999999",
          "@referredType": "Customer"
        },
        {
          "@type": "EntityRef",
          "id": "S123456789",
          "@referredType": "Service"
        },
        {
          "@type": "EntityRef",
          "id": "1-44up07",
          "@referredType": "Product"
        }
      ]
    },
    "permissionSpecificationSet": {
      "@type": "PermissionSpecificationSetRef",
      "@referredType": "PermissionSpecificationSet",
      "id": "pss-5de89f4f"
    }
  }
]
}

```

Response

```

201
{
  "@type": "PermissionSet",
  "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permission/Prms-63116b98",
  "id": "63116b98",
  "creationDate": "2023-03-01T16:49:00.000Z",
  "description": "Owner profile granted to user automatically for its owned customer, product and service entities.",
  "validFor": {
    "startDateTime": "2023-03-01T00:00:00.000Z",
    "endDateTime": "2023-04-01T00:00:00.000Z"
  },
  "user": {
    "role": "family member",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "cclt-456745",
      "href": "https://host:port/partyManagement/v5/party/cclt-456745",
      "name": "John Baker",

```

```

    "@type": "PartyRef",
    "@referredType": "Individual"
  },
  "permission": [
    {
      "@type": "Permission",
      "managedAssetGroup": {
        "@type": "ListAssetGroup",
        "@baseType": "AssetGroup",
        "entity": [
          {
            "@type": "EntityRef",
            "href": "https://serverRoot/customerManagement/v5/customer/C123456789",
            "id": "C123456789",
            "@referredType": "Customer"
          },
          {
            "@type": "EntityRef",
            "href": "https://serverRoot/customerManagement/v5/customer/5.99999.99.99.999999",
            "id": "5.99999.99.99.999999",
            "@referredType": "Customer"
          },
          {
            "@type": "EntityRef",
            "href": "https://serverRoot/serviceInventory/v5/service/S123456789",
            "id": "S123456789",
            "@referredType": "Service"
          },
          {
            "@type": "EntityRef",
            "href": "https://serverRoot/productInventory/v5/service/1-44up07",
            "id": "1-44up07",
            "@referredType": "Product"
          }
        ]
      }
    },
    {
      "permissionSpecificationSet": {
        "@type": "PermissionSpecificationSetRef",
        "@referredType": "PermissionSpecificationSet",
        "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-5de89f4f",
        "id": "pss-5de89f4f"
      }
    }
  ]
}

```

Creation of a new PermissionSet with POST operation

Request

```

POST /permissionSet
Content-Type: application/json
{
  "@type": "PermissionSet",
  "description": "Customer selfcare admin profile granted by owner to delegate
management of customers, products and services entities.",

```

```

"validFor": {
  "startDateTime": "2023-03-01T00:00:00.000Z",
  "endDateTime": "2023-04-01T00:00:00.000Z"
},
"user": {
  "role": "family member",
  "@type": "RelatedSecurityPrincipal",
  "securityPrincipal": {
    "id": "cclt-456745",
    "@type": "PartyRef",
    "@referredType": "Individual"
  }
},
"granter": {
  "role": "customer advisor",
  "@type": "RelatedPartyRefOrPartyRoleRef",
  "partyOrPartyRole": {
    "id": "p7654321",
    "@type": "PartyRef",
    "@referredType": "Individual"
  }
},
"permission": [
  {
    "@type": "Permission",
    "managedAssetGroup": {
      "@type": "ListAssetGroup",
      "@baseType": "AssetGroup",
      "entity": [
        {
          "@type": "EntityRef",
          "id": "C123456789",
          "@referredType": "Customer"
        }
      ]
    },
    "permissionSpecificationSet": {
      "@type": "PermissionSpecificationSetRef",
      "@referredType": "PermissionSpecificationSet",
      "id": "pss-edd9d8a4"
    }
  },
  {
    "@type": "Permission",
    "managedAssetGroup": {
      "@type": "ListAssetGroup",
      "@baseType": "AssetGroup",
      "entity": [
        {
          "@type": "EntityRef",
          "id": "5.99999.99.99.999999",
          "@referredType": "Customer"
        }
      ]
    },
    "permissionSpecificationSet": {
      "@type": "PermissionSpecificationSetRef",
      "@referredType": "PermissionSpecificationSet",
      "id": "pss-edd9d8a4"
    }
  },
  {
    "@type": "Permission",

```

```

    "managedAssetGroup": {
      "@type": "ListAssetGroup",
      "@baseType": "AssetGroup",
      "entity": [
        {
          "@type": "EntityRef",
          "id": "1-44up07",
          "@referredType": "Product"
        }
      ]
    },
    "permissionSpecificationSet": {
      "@type": "PermissionSpecificationSetRef",
      "@referredType": "PermissionSpecificationSet",
      "id": "pss-edd9d8a4"
    }
  },
  {
    "@type": "Permission",
    "managedAssetGroup": {
      "@type": "ListAssetGroup",
      "@baseType": "AssetGroup",
      "entity": [
        {
          "@type": "EntityRef",
          "id": "S123456789",
          "@referredType": "Service"
        }
      ]
    },
    "permissionSpecificationSet": {
      "@type": "PermissionSpecificationSetRef",
      "@referredType": "PermissionSpecificationSet",
      "id": "pss-edd9d8a4"
    }
  }
]
}

```

Response

```

201
{
  "@type": "PermissionSet",
  "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permission/Prms-7a0cb197",
  "id": "7a0cb197",
  "creationDate": "2023-03-01T16:49:00.000Z",
  "description": "Customer selfcare admin profile granted by owner to delegate management of customers, products and services entities.",
  "validFor": {
    "startDateTime": "2023-03-01T00:00:00.000Z",
    "endDateTime": "2023-04-01T00:00:00.000Z"
  },
  "user": {
    "role": "family member",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "cclt-456745",
      "href": "https://host:port/partyManagement/v5/party/cclt-456745",

```

```

    "name": "John Baker",
    "@type": "PartyRef",
    "@referredType": "Individual"
  }
},
"granter": {
  "role": "customer advisor",
  "@type": "RelatedPartyRefOrPartyRoleRef",
  "partyOrPartyRole": {
    "id": "p7654321",
    "href": "https://host:port/partyManagement/v5/party/p7654321",
    "name": "Lynda Owens",
    "@type": "PartyRef",
    "@referredType": "Individual"
  }
},
"permission": [
  {
    "@type": "Permission",
    "managedAssetGroup": {
      "@type": "ListAssetGroup",
      "@baseType": "AssetGroup",
      "entity": [
        {
          "@type": "EntityRef",
          "href": "https://serverRoot/customerManagement/v5/customer/C123456789",
          "id": "C123456789",
          "@referredType": "Customer"
        }
      ]
    }
  },
  {
    "permissionSpecificationSet": {
      "@type": "PermissionSpecificationSetRef",
      "@referredType": "PermissionSpecificationSet",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
edd9d8a4",
      "id": "pss-edd9d8a4"
    }
  },
  {
    "@type": "Permission",
    "managedAssetGroup": {
      "@type": "ListAssetGroup",
      "@baseType": "AssetGroup",
      "entity": [
        {
          "@type": "EntityRef",
          "href":
"https://serverRoot/customerManagement/v5/customer/5.99999.99.99.999999",
          "id": "5.99999.99.99.999999",
          "@referredType": "Customer"
        }
      ]
    }
  },
  {
    "permissionSpecificationSet": {
      "@type": "PermissionSpecificationSetRef",
      "@referredType": "PermissionSpecificationSet",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
edd9d8a4",
      "id": "pss-edd9d8a4"
    }
  }
]

```

```

    },
    {
      "@type": "Permission",
      "managedAssetGroup": {
        "@type": "ListAssetGroup",
        "@baseType": "AssetGroup",
        "entity": [
          {
            "@type": "EntityRef",
            "href": "https://serverRoot/productInventory/v5/service/1-44up07",
            "id": "1-44up07",
            "@referredType": "Product"
          }
        ]
      }
    },
    {
      "permissionSpecificationSet": {
        "@type": "PermissionSpecificationSetRef",
        "@referredType": "PermissionSpecificationSet",
        "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
edd9d8a4",
        "id": "pss-edd9d8a4"
      }
    },
    {
      "@type": "Permission",
      "managedAssetGroup": {
        "@type": "ListAssetGroup",
        "@baseType": "AssetGroup",
        "entity": [
          {
            "@type": "EntityRef",
            "href": "https://serverRoot/serviceInventory/v5/service/S123456789",
            "id": "S123456789",
            "@referredType": "Service"
          }
        ]
      }
    },
    {
      "permissionSpecificationSet": {
        "@type": "PermissionSpecificationSetRef",
        "@referredType": "PermissionSpecificationSet",
        "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
edd9d8a4",
        "id": "pss-edd9d8a4"
      }
    }
  ]
}

```

Creation of a new PermissionSet with POST operation

Request

```

POST /permissionSet
Content-Type: application/json
{
  "@type": "PermissionSet",
  "description": "Customer OTT primary user profile granted by owner to delegate
primary usage of an OTT service.",

```

```

"validFor": {
  "startDateTime": "2023-03-01T00:00:00.000Z",
  "endDateTime": "2023-04-01T00:00:00.000Z"
},
"user": {
  "role": "family owner",
  "@type": "RelatedSecurityPrincipal",
  "securityPrincipal": {
    "id": "cust-745712-A",
    "@type": "PartyRef",
    "@referredType": "Individual"
  }
},
"granter": {
  "role": "customer advisor",
  "@type": "RelatedPartyRefOrPartyRoleRef",
  "partyOrPartyRole": {
    "id": "cclt-456745",
    "@type": "PartyRef",
    "@referredType": "Individual"
  }
},
"permission": [
  {
    "@type": "Permission",
    "managedAssetGroup": {
      "@type": "ListAssetGroup",
      "@baseType": "AssetGroup",
      "entity": [
        {
          "@type": "EntityRef",
          "id": "S123456789",
          "@referredType": "Service"
        }
      ]
    },
    "permissionSpecificationSet": {
      "@type": "PermissionSpecificationSetRef",
      "@referredType": "PermissionSpecificationSet",
      "id": "pss-8f1a5628"
    }
  }
]
}

```

Response

```

201
{
  "@type": "PermissionSet",
  "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permission/Prms-bdfbeb06",
  "id": "bdfbeb06",
  "creationDate": "2023-03-01T16:49:00.000Z",
  "description": "Customer OTT primary user profile granted by owner to delegate primary usage of an OTT service.",
  "validFor": {
    "startDateTime": "2023-03-01T00:00:00.000Z",
    "endDateTime": "2023-04-01T00:00:00.000Z"
  },
}

```

```

"user": {
  "role": "family owner",
  "@type": "RelatedSecurityPrincipal",
  "securityPrincipal": {
    "id": "cust-745712-A",
    "href": "https://host:port/partyManagement/v5/party/cust-745712-A",
    "name": "Jack Marshall",
    "@type": "PartyRef",
    "@referredType": "Individual"
  }
},
"granter": {
  "role": "customer advisor",
  "@type": "RelatedPartyRefOrPartyRoleRef",
  "partyOrPartyRole": {
    "id": "cclt-456745",
    "href": "https://host:port/partyManagement/v5/party/cclt-456745",
    "name": "John Baker",
    "@type": "PartyRef",
    "@referredType": "Individual"
  }
},
"permission": [
  {
    "@type": "Permission",
    "managedAssetGroup": {
      "@type": "ListAssetGroup",
      "@baseType": "AssetGroup",
      "entity": [
        {
          "@type": "EntityRef",
          "href": "https://serverRoot/serviceInventory/v5/service/S123456789",
          "id": "S123456789",
          "@referredType": "Service"
        }
      ]
    }
  },
  {
    "permissionSpecificationSet": {
      "@type": "PermissionSpecificationSetRef",
      "@referredType": "PermissionSpecificationSet",
      "href":
        "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-8f1a5628",
      "id": "pss-8f1a5628"
    }
  }
]
}

```

Creation of a new PermissionSet with POST operation

Request

```

POST /permissionSet
Content-Type: application/json
{
  "@type": "PermissionSet",
  "description": "Product Catalog Management webapp TMF620 API consumer role
permission set",
  "validFor": {

```

```

    "startDateTime": "2025-10-01T00:00:00.000Z",
    "endDateTime": "2025-10-31T23:59:59.000Z"
  },
  "user": {
    "role": "tmf-api-consumer",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "rr-82288122",
      "name": "Product Catalog Management Web Application TMF Open API Consumer
Role",
      "@type": "ResourceRoleRef",
      "@referredType": "ResourceRole"
    }
  },
  "granter": {
    "role": "product_catalog_mng_admin",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "p-70fabcac9b01",
      "name": "Darth Vader",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "permission": [
    {
      "@type": "Permission",
      "permissionSpecification": {
        "@type": "PermissionSpecification",
        "name": "Catalog:ReadWrite",
        "function": "Catalog",
        "action": "ReadWrite"
      },
      "managedAssetGroup": {
        "@type": "SetAssetGroup",
        "@baseType": "AssetGroup",
        "name": "Catalog Asset Group Set",
        "entityType": [
          "Catalog"
        ]
      }
    },
    {
      "@type": "Permission",
      "permissionSpecification": {
        "@type": "PermissionSpecification",
        "name": "ProductOffering:ReadWrite",
        "function": "ProductOffering",
        "action": "ReadWrite"
      },
      "managedAssetGroup": {
        "@type": "SetAssetGroup",
        "@baseType": "AssetGroup",
        "name": "ProductOffering Asset Group Set",
        "entityType": [
          "ProductOffering"
        ]
      }
    },
    {
      "@type": "Permission",
      "permissionSpecification": {
        "@type": "PermissionSpecification",

```

```

        "name": "ProductOfferingPrice:ReadWrite",
        "function": "ProductOfferingPrice",
        "action": "ReadWrite"
    },
    "managedAssetGroup": {
        "@type": "SetAssetGroup",
        "@baseType": "AssetGroup",
        "name": "ProductOfferingPrice Asset Group Set",
        "entityType": [
            "ProductOfferingPrice"
        ]
    }
}
]
}

```

Response

```

201
{
  "@type": "PermissionSet",
  "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSet/pst-9c512f49",
  "id": "pst-9c512f49",
  "creationDate": "2025-10-01T00:00:00.000Z",
  "description": "Product Catalog Management webapp TMF620 API consumer role permission set",
  "validFor": {
    "startDateTime": "2025-10-01T00:00:00.000Z",
    "endDateTime": "2025-10-31T23:59:59.000Z"
  },
  "user": {
    "role": "tmf-api-consumer",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "rr-82288122",
      "href": "https://host:port/resourceRoleManagement/v5/resourceRole/rr-82288122",
      "name": "Product Catalog Management Web Application TMF Open API Consumer Role",
      "@type": "ResourceRoleRef",
      "@referredType": "ResourceRole"
    }
  },
  "granter": {
    "role": "product_catalog_mng_admin",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "p-70fabcac9b01",
      "href": "https://host:port/partyManagement/v5/party/p-70fabcac9b01",
      "name": "Darth Vader",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "permission": [
    {
      "@type": "Permission",
      "permissionSpecification": {
        "@type": "PermissionSpecification",

```

```

    "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-31d5b770",
    "id": "psp-31d5b770",
    "name": "Catalog:ReadWrite",
    "function": "Catalog",
    "action": "ReadWrite"
  },
  "managedAssetGroup": {
    "@type": "SetAssetGroup",
    "@baseType": "AssetGroup",
    "name": "Catalog Asset Group Set",
    "entityType": [
      "Catalog"
    ]
  }
},
{
  "@type": "Permission",
  "permissionSpecification": {
    "@type": "PermissionSpecification",
    "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-d9f344d5",
    "id": "psp-d9f344d5",
    "name": "ProductOffering:ReadWrite",
    "function": "ProductOffering",
    "action": "ReadWrite"
  },
  "managedAssetGroup": {
    "@type": "SetAssetGroup",
    "@baseType": "AssetGroup",
    "name": "ProductOffering Asset Group Set",
    "entityType": [
      "ProductOffering"
    ]
  }
},
{
  "@type": "Permission",
  "permissionSpecification": {
    "@type": "PermissionSpecification",
    "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-7fa60a0f",
    "id": "psp-7fa60a0f",
    "name": "ProductOfferingPrice:ReadWrite",
    "function": "ProductOfferingPrice",
    "action": "ReadWrite"
  },
  "managedAssetGroup": {
    "@type": "SetAssetGroup",
    "@baseType": "AssetGroup",
    "name": "ProductOfferingPrice Asset Group Set",
    "entityType": [
      "ProductOfferingPrice"
    ]
  }
}
]
}

```

Updates partially a PermissionSet

PATCH /permissionSet/{id}?fields=...

Description

This operation allows partial updates of a PermissionSet entity. Support of json/merge (<https://tools.ietf.org/html/rfc7396>) is mandatory, support of json/patch (<http://tools.ietf.org/html/rfc5789>) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
creationDate	
href	
id	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
description	
granter	
permission	
permissionSetSpecification	
user	
validFor	

Usage samples

Example of a request for partial updating a permission set, using JSON Patch.

Request

```
PATCH /permissionSet/Prms-jnzgh456
Content-Type: application/json-patch+json
[
  {
    "op": "replace",
    "path": "/validFor/endTime",
    "value": "2021-11-31T23:59:59.000Z"
  }
]
```

Response

```

200
{
  "@type": "PermissionSet",
  "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSet/Prms-jnzgh456",
  "id": "Prms-jnzgh456",
  "creationDate": "2019-10-01T00:00:00.000Z",
  "description": "superProfile Granted for user",
  "validFor": {
    "startTime": "2019-10-01T00:00:00.000Z",
    "endTime": "2021-10-31T23:59:59.000Z"
  },
  "user": {
    "role": "client",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "cust-745712-A",
      "href": "https://host:port/partyManagement/v5/party/cust-745712-A",
      "name": "Jack Marshall",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "granter": {
    "role": "customer advisor",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "cclt-456745",
      "href": "https://host:port/partyManagement/v5/party/cclt-456745",
      "name": "John Baker",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "permission": [
    {
      "@type": "Permission",
      "permissionSpecification": {
        "@type": "PermissionSpecification",
        "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-31d5b770",
        "id": "ps-31d5b770",
        "function": "ProductOrder",
        "action": "update"
      },
      "managedAssetGroup": {
        "@type": "FilterAssetGroup",
        "@baseType": "AssetGroup",
        "entityType": "ProductOrder",
        "filter": "relatedParty.id=\"cid3366116688\"&relatedParty.@referredType=\"Customer\""
      }
    }
  ]
}

```

Example of a request for partial updating a permission set, using MERGE Patch.

Request

```
PATCH /permissionSet/Prms-jnzgh456
Content-Type: application/merge-patch+json
{
  "@type": "PermissionSet",
  "validFor": {
    "startDateTime": "2021-10-01T00:00:00.000Z",
    "endDateTime": "2023-10-31T23:59:59.000Z"
  }
}
```

Response

```
200
{
  "@type": "PermissionSet",
  "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSet/Prms-jnzgh456",
  "id": "Prms-jnzgh456",
  "creationDate": "2019-10-01T00:00:00.000Z",
  "description": "superProfile Granted for user",
  "validFor": {
    "startDateTime": "2021-10-01T00:00:00.000Z",
    "endDateTime": "2023-10-31T23:59:59.000Z"
  },
  "user": {
    "role": "client",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "cust-745712-A",
      "href": "https://host:port/partyManagement/v5/party/cust-745712-A",
      "name": "Jack Marshall",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "granter": {
    "role": "customer advisor",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "cclt-456745",
      "href": "https://host:port/partyManagement/v5/party/cclt-456745",
      "name": "John Baker",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "permission": [
    {
      "@type": "Permission",
      "permissionSpecification": {
        "@type": "PermissionSpecification",
        "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-31d5b770",
        "id": "ps-31d5b770",
        "function": "ProductOrder",
        "action": "update"
      }
    }
  ]
}
```

```

    },
    "managedAssetGroup": {
      "@type": "FilterAssetGroup",
      "@baseType": "AssetGroup",
      "entityType": "ProductOrder",
      "filter":
"relatedParty.id=\"cid3366116688\"&relatedParty.@referredType=\"Customer\""
    }
  }
]
}

```

Example of a request for fully updating an permission set identity.

Request

```

PATCH /permissionSet/Prms-jnzgh456
Content-Type: application/json
{
  "@type": "PermissionSet",
  "description": "superProfile Granted for user",
  "validFor": {
    "startDateTime": "2021-10-01T00:00:00.000Z",
    "endDateTime": "2023-10-31T23:59:59.000Z"
  },
  "user": {
    "role": "client",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "cust-745712-A",
      "name": "Jack Marshall",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "granter": {
    "role": "customer advisor",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "cclt-456745",
      "name": "John Baker",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "permission": [
    {
      "@type": "Permission",
      "permissionSpecification": {
        "@type": "PermissionSpecification",
        "name": "ProductOrder:update",
        "function": "ProductOrder",
        "action": "update"
      },
      "managedAssetGroup": {
        "@type": "FilterAssetGroup",
        "@baseType": "AssetGroup",
        "entityType": "ProductOrder",
        "filter":
"relatedParty.id=\"cid3366116688\"&relatedParty.@referredType=\"Customer\""
    }
  ]
}

```

```

    }
  }
}

```

Response

```

200
{
  "@type": "PermissionSet",
  "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSet/Prms-jnzgh456",
  "id": "Prms-jnzgh456",
  "creationDate": "2019-10-01T00:00:00.000Z",
  "description": "superProfile Granted for user",
  "validFor": {
    "startDateTime": "2021-10-01T00:00:00.000Z",
    "endDateTime": "2023-10-31T23:59:59.000Z"
  },
  "user": {
    "role": "client",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "cust-745712-A",
      "href": "https://host:port/partyManagement/v5/party/cust-745712-A",
      "name": "Jack Marshall",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "granter": {
    "role": "customer advisor",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "cclt-456745",
      "href": "https://host:port/partyManagement/v5/party/cclt-456745",
      "name": "John Baker",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "permission": [
    {
      "@type": "Permission",
      "permissionSpecification": {
        "@type": "PermissionSpecification",
        "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-31d5b770",
        "id": "ps-31d5b770",
        "function": "ProductOrder",
        "action": "update"
      },
      "managedAssetGroup": {
        "@type": "FilterAssetGroup",
        "@baseType": "AssetGroup",
        "entityType": "ProductOrder",
        "filter": "relatedParty.id=\"cid3366116688\"&relatedParty.@referredType=\"Customer\""
      }
    }
  ]
}

```

```

    }
  ]
}

```

Example of a request for partial updating a permission set, using JSON Patch Query.

Request

```

PATCH /permissionSet/413920
Content-Type: application/json-patch-query+json
[
  {
    "op": "remove",
    "path":
"/permission/managedAssetGroup?permission.managedAssetGroup.entityType=ProductOrder"
  }
]

```

Response

```

200
{
  "@type": "PermissionSet",
  "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSet/Prms-jnzgh456",
  "id": "Prms-jnzgh456",
  "creationDate": "2019-10-01T00:00:00.000Z",
  "description": "superProfile Granted for user",
  "validFor": {
    "startTime": "2019-10-01T00:00:00.000Z",
    "endTime": "2021-10-31T23:59:59.000Z"
  },
  "user": {
    "role": "client",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "cust-745712-A",
      "href": "https://host:port/partyManagement/v5/party/cust-745712-A",
      "name": "Jack Marshall",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "granter": {
    "role": "customer advisor",
    "@type": "RelatedPartyRefOrPartyRoleRef",
    "partyOrPartyRole": {
      "id": "cclt-456745",
      "href": "https://host:port/partyManagement/v5/party/cclt-456745",
      "name": "John Baker",
      "@type": "PartyRef",
      "@referredType": "Individual"
    }
  },
  "permission": [
    {
      "@type": "Permission",

```

```
    "permissionSpecification": {
      "@type": "PermissionSpecification",
      "href":
        "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-31d5b770",
      "id": "ps-31d5b770",
      "function": "ProductOrder",
      "action": "update"
    }
  }
]
```

Deletes a PermissionSet

DELETE /permissionSet/{id}

Description

This operation deletes a PermissionSet entity.

Usage samples

Here's an example of a request for delete a specific permission set.

Request

```
DELETE /permissionSet/07252355872483
Content-Type: application/json
```

Response

204

Operations on PermissionSpecificationSet

Retrieves a PermissionSpecificationSet by ID

GET /permissionSpecificationSet/{id}?fields=...

Description

This operation retrieves a PermissionSpecificationSet entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a permission specification set by its identifier

Request

```
GET /permissionSpecificationSet/pss-ebe41def
Content-Type: application/json
```

Response

```
200
{
  "@type": "PermissionSpecificationSet",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
  ebe41def",
  "id": "pss-ebe41def",
  "involvementRole": "MyTV familyAdmin Profile",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecificationRef",
      "@referredType": "PermissionSpecification",
      "href":
      "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
      5afe52c5",
      "id": "pss-5afe52c5"
    },
    {
      "@type": "PermissionSpecificationRef",
      "@referredType": "PermissionSpecification",
      "href":
      "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
      769ef409",
      "id": "pss-769ef409"
    }
  ]
}
```

Retrieve a permission specification set by its identifier

Request

```
GET /permissionSpecificationSet/pss-5de89f4f
Content-Type: application/json
```

Response

```
200
{
  "@type": "PermissionSpecificationSet",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
  5de89f4f",
  "id": "pss-5de89f4f",
  "involvementRole": "Owner",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecification",
      "href":

```

```

"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-9d5692ea",
  "id": "ps-9d5692ea",
  "name": "all:readwrite",
  "function": "all",
  "action": "ReadWrite"
}
]
}

```

Retrieve a permission specification set by its identifier

Request

```

GET /permissionSpecificationSet/pss-edd9d8a4
Content-Type: application/json

```

Response

```

200
{
  "@type": "PermissionSpecificationSet",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-edd9d8a4",
  "id": "pss-edd9d8a4",
  "involvementRole": "Customer.Selfcare.Admin",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecification",
      "href":
      "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-dd798b34",
      "id": "ps-dd798b34",
      "description": "Permission to manage users and to add or remove permissions to delegated users.",
      "name": "usermanagement:readwrite",
      "function": "user_management",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "href":
      "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-591c77a2",
      "id": "ps-591c77a2",
      "description": "Permission to view and change account data details.",
      "name": "accountdetails:readwrite",
      "function": "account_details",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "href":
      "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-9f32c9e0",
      "id": "ps-9f32c9e0",

```

```

    "description": "Permission to place request to change holder data.",
    "name": "holderdata:change",
    "function": "change_holder_data",
    "action": "ReadWrite",
    "specificationCharacteristic": [
      {
        "@type": "CharacteristicSpecification",
        "name": "Macro.Segment",
        "valueType": "string",
        "characteristicValueSpecification": [
          {
            "@type": "StringCharacteristicValueSpecification",
            "valueType": "string",
            "value": "mass_market"
          },
          {
            "@type": "StringCharacteristicValueSpecification",
            "valueType": "string",
            "value": "residential"
          }
        ]
      }
    ]
  }
]
}

```

Retrieve a permission specification set by its identifier

Request

```

GET /permissionSpecificationSet/pss-8f1a5628
Content-Type: application/json

```

Response

```

200
{
  "@type": "PermissionSpecificationSet",
  "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-8f1a5628",
  "id": "pss-8f1a5628",
  "involvementRole": "Customer.OTT.Primary.User",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecification",
      "href":
        "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-a99c47c6",
      "id": "ps-a99c47c6",
      "description": "Permission to manage users and to add or remove permissions to delegated users.",
      "name": "usermanagement:readwrite",
      "function": "user_management",
      "action": "ReadWrite"
    },
  ],
}

```

```
{
  "@type": "PermissionSpecification",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-5ea0f3cc",
  "id": "ps-5ea0f3cc",
  "description": "Permission to buy products in OTT apps.",
  "name": "impulsebuy:enabled",
  "function": "impulse_buy",
  "action": "Enabled"
},
{
  "@type": "PermissionSpecification",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-b4b3601a",
  "id": "ps-b4b3601a",
  "description": "Permission to manage delegated users parental controls.",
  "name": "parentalcontrols:manage",
  "function": "manage_parental_controls",
  "action": "ReadWrite"
}
]
```

Retrieve a permission specification set by its identifier

Request

```
GET /permissionSpecificationSet/pss-8f1a5628
Content-Type: application/json
```

Response

```
200
{
  "@type": "PermissionSpecificationSet",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-9f346b96",
  "id": "pss-9f346b96",
  "involvementRole": "TMF620 API Provider Role",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecification",
      "href":
      "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-31d5b770",
      "id": "psp-31d5b770",
      "name": "Catalog:ReadWrite",
      "function": "Catalog",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "href":
      "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-
```

```

d9f344d5",
  "id": "psp-d9f344d5",
  "name": "ProductOffering:ReadWrite",
  "function": "ProductOffering",
  "action": "ReadWrite"
},
{
  "@type": "PermissionSpecification",
  "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-
7fa60a0f",
  "id": "psp-7fa60a0f",
  "name": "ProductOfferingPrice:ReadWrite",
  "function": "ProductOfferingPrice",
  "action": "ReadWrite"
},
{
  "@type": "PermissionSpecification",
  "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-
0bd79203",
  "id": "psp-0bd79203",
  "name": "ProductSpecification:ReadWrite",
  "function": "ProductSpecification",
  "action": "ReadWrite"
},
{
  "@type": "PermissionSpecification",
  "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-
873b7ab5",
  "id": "psp-873b7ab5",
  "name": "Category:ReadWrite",
  "function": "Category",
  "action": "ReadWrite"
}
]
}

```

List or find PermissionSpecificationSet objects

GET /permissionSpecificationSet?fields=...

Description

This operation list PermissionSpecificationSet entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a list of permission specification sets

Request

```

GET /permissionSpecificationSet
Content-Type: application/json

```

Response

```

200
[
  {
    "@type": "PermissionSpecificationSet",
    "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
    ebe41def",
    "id": "pss-ebe41def",
    "involvementRole": "MyTV familyAdmin Profile",
    "permissionSpecification": [
      {
        "@type": "PermissionSpecificationRef",
        "@referredType": "PermissionSpecification",
        "href":
        "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
        5afe52c5",
        "id": "pss-5afe52c5"
      },
      {
        "@type": "PermissionSpecificationRef",
        "@referredType": "PermissionSpecification",
        "href":
        "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
        769ef409",
        "id": "pss-769ef409"
      }
    ]
  },
  {
    "@type": "PermissionSpecificationSet",
    "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
    5de89f4f",
    "id": "pss-5de89f4f",
    "involvementRole": "Owner",
    "permissionSpecification": [
      {
        "@type": "PermissionSpecification",
        "href":
        "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
        9d5692ea",
        "id": "ps-9d5692ea",
        "name": "all:readwrite",
        "function": "all",
        "action": "ReadWrite"
      }
    ]
  },
  {
    "@type": "PermissionSpecificationSet",
    "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
    edd9d8a4",
    "id": "pss-edd9d8a4",
    "involvementRole": "Customer.Selfcare.Admin",
    "permissionSpecification": [
      {
        "@type": "PermissionSpecification",
        "href":

```

```

"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
dd798b34",
  "id": "ps-dd798b34",
  "description": "Permission to manage users and to add or remove permissions
to delegated users.",
  "name": "usermanagement:readwrite",
  "function": "user_management",
  "action": "ReadWrite"
},
{
  "@type": "PermissionSpecification",
  "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
591c77a2",
  "id": "ps-591c77a2",
  "description": "Permission to view and change account data details.",
  "name": "accountdetails:readwrite",
  "function": "account_details",
  "action": "ReadWrite"
},
{
  "@type": "PermissionSpecification",
  "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
9f32c9e0",
  "id": "ps-9f32c9e0",
  "description": "Permission to place request to change holder data.",
  "name": "holderdata:change",
  "function": "change_holder_data",
  "action": "ReadWrite",
  "specificationCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "Macro.Segment",
      "valueType": "string",
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "mass_market"
        },
        {
          "@type": "StringCharacteristicValueSpecification",
          "valueType": "string",
          "value": "residential"
        }
      ]
    }
  ]
},
{
  "@type": "PermissionSpecificationSet",
  "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
8f1a5628",
  "id": "pss-8f1a5628",
  "involvementRole": "Customer.OTT.Primary.User",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecification",
      "href":

```

```

"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
a99c47c6",
  "id": "ps-a99c47c6",
  "description": "Permission to manage users and to add or remove permissions
to delegated users.",
  "name": "usermanagement:readwrite",
  "function": "user_management",
  "action": "ReadWrite"
},
{
  "@type": "PermissionSpecification",
  "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
5ea0f3cc",
  "id": "ps-5ea0f3cc",
  "description": "Permission to buy products in OTT apps.",
  "name": "impulsebuy:enabled",
  "function": "impulse_buy",
  "action": "Enabled"
},
{
  "@type": "PermissionSpecification",
  "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
b4b3601a",
  "id": "ps-b4b3601a",
  "description": "Permission to manage delegated users parental controls.",
  "name": "parentalcontrols:manage",
  "function": "manage_parental_controls",
  "action": "ReadWrite"
}
]
},
{
  "@type": "PermissionSpecificationSet",
  "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-
9f346b96",
  "id": "pss-9f346b96",
  "involvementRole": "TMF620 API Provider Role",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecification",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-
31d5b770",
      "id": "psp-31d5b770",
      "name": "Catalog:ReadWrite",
      "function": "Catalog",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-
d9f344d5",
      "id": "psp-d9f344d5",
      "name": "ProductOffering:ReadWrite",
      "function": "ProductOffering",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",

```

```

      "href":
      "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-7fa60a0f",
      "id": "psp-7fa60a0f",
      "name": "ProductOfferingPrice:ReadWrite",
      "function": "ProductOfferingPrice",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "href":
      "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-0bd79203",
      "id": "psp-0bd79203",
      "name": "ProductSpecification:ReadWrite",
      "function": "ProductSpecification",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "href":
      "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-873b7ab5",
      "id": "psp-873b7ab5",
      "name": "Category:ReadWrite",
      "function": "Category",
      "action": "ReadWrite"
    }
  ]
}
]

```

Creates a PermissionSpecificationSet

POST /permissionSpecificationSet?fields=...

Description

This operation creates a PermissionSpecificationSet entity.

Mandatory Attributes

Mandatory Attributes	Rule
name	
permissionSpecification	
@type	

Usage samples

Creation of a new PermissionSpecificationSet with POST operation

Request

```

POST /permissionSpecificationSet
Content-Type: application/json

```

```
{
  "@type": "PermissionSpecificationSet",
  "name": "Customer.OTT.Primary.User",
  "involvementRole": "Customer.OTT.Primary.User",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecification",
      "description": "Permission to manage users and to add or remove permissions to delegated users.",
      "name": "usermanagement:readwrite",
      "function": "user_management",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "description": "Permission to buy products in OTT apps.",
      "name": "impulsebuy:enabled",
      "function": "impulse_buy",
      "action": "Enabled"
    },
    {
      "@type": "PermissionSpecification",
      "description": "Permission to manage delegated users parental controls.",
      "name": "parentalcontrols:manage",
      "function": "manage_parental_controls",
      "action": "ReadWrite"
    }
  ]
}
```

Response

```
201
{
  "@type": "PermissionSpecificationSet",
  "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/pss-8f1a5628",
  "id": "pss-8f1a5628",
  "name": "Customer.OTT.Primary.User",
  "involvementRole": "Customer.OTT.Primary.User",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecification",
      "href":
        "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-a99c47c6",
      "id": "ps-a99c47c6",
      "description": "Permission to manage users and to add or remove permissions to delegated users.",
      "name": "usermanagement:readwrite",
      "function": "user_management",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "href":
        "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-5ea0f3cc",
      "id": "ps-5ea0f3cc",

```

```

    "description": "Permission to buy products in OTT apps.",
    "name": "impulsebuy:enabled",
    "function": "impulse_buy",
    "action": "Enabled"
  },
  {
    "@type": "PermissionSpecification",
    "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
b4b3601a",
    "id": "ps-b4b3601a",
    "description": "Permission to manage delegated users parental controls.",
    "name": "parentalcontrols:manage",
    "function": "manage_parental_controls",
    "action": "ReadWrite"
  }
]
}

```

Creation of a new PermissionSpecificationSet with POST operation

Request

```

POST /permissionSpecificationSet
Content-Type: application/json
{
  "@type": "PermissionSpecificationSet",
  "name": "TMF620 API Provider Role",
  "involvementRole": "TMF620 API Provider Role",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecification",
      "name": "Catalog:ReadWrite",
      "function": "Catalog",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "name": "ProductOffering:ReadWrite",
      "function": "ProductOffering",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "name": "ProductOfferingPrice:ReadWrite",
      "function": "ProductOfferingPrice",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "name": "ProductSpecification:ReadWrite",
      "function": "ProductSpecification",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "name": "Category:ReadWrite",
      "function": "Category",
      "action": "ReadWrite"
    }
  ]
}

```

```
]
}
```

Response

```
201
{
  "@type": "PermissionSpecificationSet",
  "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/ps-
    9f346b96",
  "id": "pss-9f346b96",
  "involvementRole": "TMF620 API Provider Role",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecification",
      "href":
        "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-
        31d5b770",
      "id": "psp-31d5b770",
      "name": "Catalog:ReadWrite",
      "function": "Catalog",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "href":
        "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-
        d9f344d5",
      "id": "psp-d9f344d5",
      "name": "ProductOffering:ReadWrite",
      "function": "ProductOffering",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "href":
        "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-
        7fa60a0f",
      "id": "psp-7fa60a0f",
      "name": "ProductOfferingPrice:ReadWrite",
      "function": "ProductOfferingPrice",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "href":
        "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-
        0bd79203",
      "id": "psp-0bd79203",
      "name": "ProductSpecification:ReadWrite",
      "function": "ProductSpecification",
      "action": "ReadWrite"
    },
    {
      "@type": "PermissionSpecification",
      "href":
        "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/psp-
        873b7ab5",
      "id": "psp-873b7ab5",

```

```

    "name": "Category:ReadWrite",
    "function": "Category",
    "action": "ReadWrite"
  }
]
}

```

Updates partially a PermissionSpecificationSet

PATCH /permissionSpecificationSet/{id}?fields=...

Description

This operation allows partial updates of a PermissionSpecificationSet entity. Support of json/merge (<https://tools.ietf.org/html/rfc7396>) is mandatory, support of json/patch (<http://tools.ietf.org/html/rfc5789>) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
description	
href	
id	
involvementRole	
name	
permissionSpecification	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
----------------------	------

Usage samples

Example of a request for partial updating a permission specification set, using JSON Patch Query.

Request

```

PATCH /permissionSpecificationSet/413920
Content-Type: application/json-patch-query+json
[
  {
    "op": "remove",
    "path": "/permissionSpecification?permissionSpecification.id=632514"
  }
]

```

```
}
]
```

Response

```
200
{
  "@type": "PermissionSpecificationSet",
  "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/413920",
  "id": "413920",
  "involvementRole": "MyTV SPORT+ familyAdmin Profile",
  "name": "TV Sport and Family Admin",
  "description": "Gives permissions to use Sport Channels and manage the family",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecificationRef",
      "@referredType": "PermissionSpecification",
      "href":
        "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/12014",
      "id": "12014"
    },
    {
      "@type": "PermissionSpecificationRef",
      "@referredType": "PermissionSpecification",
      "href":
        "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/960124"
    },
    {
      "@type": "PermissionSpecificationRef",
      "@referredType": "PermissionSpecification",
      "href":
        "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/6888633"
    },
    {
      "id": "6888633"
    }
  ]
}
```

Example of a request for fully updating an entire permission specification set.

Request

```
PATCH /permissionSpecificationSet/413920
Content-Type: application/json
{
  "@type": "PermissionSpecificationSet",
  "involvementRole": "MyTV SPORT+ familyAdmin Profile",
  "name": "TV Sport and Family Admin",
  "description": "Gives permissions to use Sport Channels and manage the family",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecificationRef",
      "@referredType": "PermissionSpecification",
```

```

        "id": "632514"
      },
      {
        "@type": "PermissionSpecificationRef",
        "@referredType": "PermissionSpecification",
        "id": "12014"
      },
      {
        "@type": "PermissionSpecificationRef",
        "@referredType": "PermissionSpecification",
        "id": "960124"
      },
      {
        "@type": "PermissionSpecificationRef",
        "@referredType": "PermissionSpecification",
        "id": "6888633"
      }
    ]
  }
}

```

Response

```

200
{
  "@type": "PermissionSpecificationSet",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/4139
20",
  "id": "413920",
  "involvementRole": "MyTV SPORT+ familyAdmin Profile",
  "name": "TV Sport and Family Admin",
  "description": "Gives permissions to use Sport Channels and manage the family",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecificationRef",
      "@referredType": "PermissionSpecification",
      "href":
      "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/632514"
    },
    {
      "@type": "PermissionSpecificationRef",
      "@referredType": "PermissionSpecification",
      "href":
      "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/12014",
      "id": "12014"
    },
    {
      "@type": "PermissionSpecificationRef",
      "@referredType": "PermissionSpecification",
      "href":
      "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/960124"
    },
    {
      "@type": "PermissionSpecificationRef",
      "@referredType": "PermissionSpecification",
      "href":

```

```

"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/6888633
",
  "id": "6888633"
}
]
}

```

Example of a request for partial updating a permission specification set, using MERGE Patch.

Request

```

PATCH /permissionSpecificationSet/413920
Content-Type: application/merge-patch+json
{
  "@type": "PermissionSpecificationSet",
  "name": "TV Sport and Family Admin",
  "description": "Gives permissions to use Sport Channels and manage the family"
}

```

Response

```

200
{
  "@type": "PermissionSpecificationSet",
  "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/4139
20",
  "id": "413920",
  "involvementRole": "MyTV SPORT+ familyAdmin Profile",
  "name": "TV Sport and Family Admin",
  "description": "Gives permissions to use Sport Channels and manage the family",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecificationRef",
      "@referredType": "PermissionSpecification",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/632514"
    },
    {
      "id": "632514"
    },
    {
      "@type": "PermissionSpecificationRef",
      "@referredType": "PermissionSpecification",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/12014",
      "id": "12014"
    },
    {
      "@type": "PermissionSpecificationRef",
      "@referredType": "PermissionSpecification",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/960124"
    },
    {
      "id": "960124"
    },
    {

```

```

      "@type": "PermissionSpecificationRef",
      "@referredType": "PermissionSpecification",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/6888633
",
      "id": "6888633"
    }
  ]
}

```

Example of a request for partial updating a permission specification set, using JSON Patch.

Request

```

PATCH /permissionSpecificationSet/413920
Content-Type: application/json-patch+json
[
  {
    "op": "add",
    "path": "/permissionSpecification",
    "value": {
      "id": "6888633",
      "@type": "PermissionSpecificationRef",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/6888633
"
    }
  }
]

```

Response

```

200
{
  "@type": "PermissionSpecificationSet",
  "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecificationSet/4139
20",
  "id": "413920",
  "involvementRole": "MyTV SPORT+ familyAdmin Profile",
  "permissionSpecification": [
    {
      "@type": "PermissionSpecificationRef",
      "@referredType": "PermissionSpecification",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/632514"
,
      "id": "632514"
    },
    {
      "@type": "PermissionSpecificationRef",
      "@referredType": "PermissionSpecification",
      "href":
"https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/12014",
      "id": "12014"
    },
    {

```

```
{
  "@type": "PermissionSpecificationRef",
  "@referredType": "PermissionSpecification",
  "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/960124"
  ,
  "id": "960124"
},
{
  "@type": "PermissionSpecificationRef",
  "@referredType": "PermissionSpecification",
  "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/6888633"
  ,
  "id": "6888633"
}
]
```

Deletes a PermissionSpecificationSet

DELETE /permissionSpecificationSet/{id}

Description

This operation deletes a PermissionSpecificationSet entity.

Usage samples

Here's an example of a request for delete a specific permission specification set.

Request

```
DELETE /permissionSpecificationSet/3267623312453
Content-Type: application/json
```

Response

204

Operations on PermissionSpecification

Retrieves a PermissionSpecification by ID

GET /permissionSpecification/{id}?fields=...

Description

This operation retrieves a PermissionSpecification entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a permission specification by its identifier

Request

```
GET /permissionSpecification/ps-7e0a2d18
Content-Type: application/json
```

Response

```
200
{
  "@type": "PermissionSpecification",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-7e0a2d18",
  "id": "ps-7e0a2d18",
  "name": "ProductOrder:grantAlteration",
  "function": "ProductOrder",
  "action": "grantAlteration",
  "specificationCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "reductionPercentage",
      "valueType": "number",
      "characteristicValueSpecification": [
        {
          "@type": "FloatCharacteristicValueSpecification",
          "valueType": "number",
          "value": 5.5
        }
      ]
    }
  ]
}
```

Retrieve a permission specification by its identifier

Request

```
GET /permissionSpecification/ps-7e0a2d18
Content-Type: application/json
```

Response

```
200
{
  "@type": "PermissionSpecification",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-713e2adc",
  "id": "ps-713e2adc",
  "name": "all:readwrite",
  "function": "all",
  "action": "ReadWrite"
```

```
}
```

Retrieve a permission specification by its identifier

Request

```
GET /permissionSpecification/ps-7e0a2d18
Content-Type: application/json
```

Response

```
200
{
  "@type": "PermissionSpecification",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-591c77a2",
  "id": "ps-591c77a2",
  "description": "Permission to view and change account data details.",
  "name": "accountdetails:readwrite",
  "function": "account_details",
  "action": "ReadWrite"
}
```

Retrieve a permission specification by its identifier

Request

```
GET /permissionSpecification/ps-7e0a2d18
Content-Type: application/json
```

Response

```
200
{
  "@type": "PermissionSpecification",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-9f32c9e0",
  "id": "ps-9f32c9e0",
  "description": "Permission to place request to change holder data.",
  "name": "holderdata:change",
  "function": "change_holder_data",
  "action": "ReadWrite",
  "specificationCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "Macro.Segment",
      "valueType": "string",
      "characteristicValueSpecification": [
        {
          "@type": "StringCharacteristicValueSpecification",
```

```

        "valueType": "string",
        "value": "mass_market"
      },
      {
        "@type": "StringCharacteristicValueSpecification",
        "valueType": "string",
        "value": "residential"
      }
    ]
  }
}

```

Retrieve a permission specification by its identifier

Request

```

GET /permissionSpecification/ps-7e0a2d18
Content-Type: application/json

```

Response

```

200
{
  "@type": "PermissionSpecification",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
a99c47c6",
  "id": "ps-a99c47c6",
  "description": "Permission to manage users and to add or remove permissions to
delegated users.",
  "function": "user_management",
  "action": "ReadWrite"
}

```

Retrieve a permission specification by its identifier

Request

```

GET /permissionSpecification/ps-7e0a2d18
Content-Type: application/json

```

Response

```

200
{
  "@type": "PermissionSpecification",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
5ea0f3cc",
  "id": "ps-5ea0f3cc",
  "description": "Permission to buy products in OTT apps.",

```

```
"function": "impulse_buy",  
"action": "Enabled"  
}
```

Retrieve a permission specification by its identifier

Request

```
GET /permissionSpecification/ps-7e0a2d18  
Content-Type: application/json
```

Response

```
200  
{  
  "@type": "PermissionSpecification",  
  "href":  
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-  
b4b3601a",  
  "id": "ps-b4b3601a",  
  "description": "Permission to manage delegated users parental controls.",  
  "function": "manage_parental_controls",  
  "action": "ReadWrite"  
}
```

List or find PermissionSpecification objects

GET /permissionSpecification?fields=...

Description

This operation list PermissionSpecification entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a list of permission specifications

Request

```
GET /permissionSpecification  
Content-Type: application/json
```

Response

```
200  
[  
  {  
    "@type": "PermissionSpecification",
```

```

    "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
    7e0a2d18",
    "id": "ps-7e0a2d18",
    "name": "ProductOrder:grantAlteration",
    "function": "ProductOrder",
    "action": "grantAlteration",
    "specificationCharacteristic": [
      {
        "@type": "CharacteristicSpecification",
        "name": "reductionPercentage",
        "valueType": "number",
        "characteristicValueSpecification": [
          {
            "@type": "FloatCharacteristicValueSpecification",
            "valueType": "number",
            "value": 5.5
          }
        ]
      }
    ]
  },
  {
    "@type": "PermissionSpecification",
    "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
    713e2adc",
    "id": "ps-713e2adc",
    "name": "all:readwrite",
    "function": "all",
    "action": "ReadWrite"
  },
  {
    "@type": "PermissionSpecification",
    "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
    591c77a2",
    "id": "ps-591c77a2",
    "description": "Permission to view and change account data details.",
    "name": "accountdetails:readwrite",
    "function": "account_details",
    "action": "ReadWrite"
  },
  {
    "@type": "PermissionSpecification",
    "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
    9f32c9e0",
    "id": "ps-9f32c9e0",
    "description": "Permission to place request to change holder data.",
    "name": "holderdata:change",
    "function": "change_holder_data",
    "action": "ReadWrite",
    "specificationCharacteristic": [
      {
        "@type": "CharacteristicSpecification",
        "name": "Macro.Segment",
        "valueType": "string",
        "characteristicValueSpecification": [
          {
            "@type": "StringCharacteristicValueSpecification",
            "valueType": "string",
            "value": "mass_market"
          }
        ]
      }
    ]
  }
]

```

```

    },
    {
      "@type": "StringCharacteristicValueSpecification",
      "valueType": "string",
      "value": "residential"
    }
  ]
}
]
},
{
  "@type": "PermissionSpecification",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
  a99c47c6",
  "id": "ps-a99c47c6",
  "description": "Permission to manage users and to add or remove permissions to
  delegated users.",
  "function": "user_management",
  "action": "ReadWrite"
},
{
  "@type": "PermissionSpecification",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
  5ea0f3cc",
  "id": "ps-5ea0f3cc",
  "description": "Permission to buy products in OTT apps.",
  "function": "impulse_buy",
  "action": "Enabled"
},
{
  "@type": "PermissionSpecification",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-
  b4b3601a",
  "id": "ps-b4b3601a",
  "description": "Permission to manage delegated users parental controls.",
  "function": "manage_parental_controls",
  "action": "ReadWrite"
}
]

```

Creates a PermissionSpecification

POST /permissionSpecification?fields=...

Description

This operation creates a PermissionSpecification entity.

Mandatory Attributes

Mandatory Attributes	Rule
action	
function	
name	M

Mandatory Attributes	Rule
@type	

Usage samples

Creation of a new PermissionSpecification with POST operation

Request

```
POST /permissionSpecification
Content-Type: application/json
{
  "@type": "PermissionSpecification",
  "function": "ProductOrder",
  "action": "grantAlteration",
  "name": "Allow adhoc price alteration",
  "specificationCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "reductionPercentage",
      "valueType": "number",
      "characteristicValueSpecification": [
        {
          "@type": "FloatCharacteristicValueSpecification",
          "valueType": "number",
          "value": 5.5
        }
      ]
    }
  ]
}
```

Response

```
201
{
  "@type": "PermissionSpecification",
  "href": "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-7e0a2d18",
  "id": "ps-7e0a2d18",
  "function": "ProductOrder",
  "action": "grantAlteration",
  "name": "Allow adhoc price alteration",
  "specificationCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "reductionPercentage",
      "valueType": "number",
      "characteristicValueSpecification": [
        {
          "@type": "FloatCharacteristicValueSpecification",
          "valueType": "number",
          "value": 5.5
        }
      ]
    }
  ]
}
```

```
]
}
```

Updates partially a PermissionSpecification

PATCH /permissionSpecification/{id}?fields=...

Description

This operation allows partial updates of a PermissionSpecification entity. Support of json/merge (<https://tools.ietf.org/html/rfc7396>) is mandatory, support of json/patch (<http://tools.ietf.org/html/rfc5789>) is optional. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
action	
description	
function	
href	
id	
name	
specificationCharacteristic	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
----------------------	------

Usage samples

Example of a request for partial updating a permission specification, using JSON Patch.

Request

```
PATCH /permissionSpecification/ps-7e0a2d18
Content-Type: application/json-patch+json
[
  {
    "op": "add",
    "path": "/specificationCharacteristic",
    "value": {
      "@type": "CharacteristicSpecification",
      "valueType": "boolean",
```

```

    "name": "reductionPercentageRestricted",
    "characteristicValueSpecification": [
      {
        "@type": "BooleanCharacteristicValueSpecification",
        "valueType": "boolean",
        "value": true
      }
    ]
  }
}
]

```

Response

```

200
{
  "@type": "PermissionSpecification",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-7e0a2d18",
  "id": "ps-7e0a2d18",
  "name": "ProductOrder:grantAlteration",
  "function": "ProductOrder",
  "action": "grantAlteration",
  "specificationCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "reductionPercentage",
      "valueType": "number",
      "characteristicValueSpecification": [
        {
          "@type": "FloatCharacteristicValueSpecification",
          "valueType": "number",
          "value": 5.5
        }
      ]
    },
    {
      "@type": "CharacteristicSpecification",
      "valueType": "boolean",
      "name": "reductionPercentageRestricted",
      "characteristicValueSpecification": [
        {
          "@type": "BooleanCharacteristicValueSpecification",
          "valueType": "boolean",
          "value": true
        }
      ]
    }
  ]
}

```

Example of a request for partial updating a permission specification, using MERGE Patch.

Request

```
PATCH /permissionSpecification/ps-7e0a2d18
```

```
Content-Type: application/merge-patch+json
{
  "@type": "PermissionSpecification",
  "specificationCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "reductionPercentage",
      "valueType": "number",
      "characteristicValueSpecification": [
        {
          "@type": "FloatCharacteristicValueSpecification",
          "valueType": "number",
          "value": 5.5
        }
      ]
    }
  ]
}
```

Response

```
200
{
  "@type": "PermissionSpecification",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-7e0a2d18",
  "id": "ps-7e0a2d18",
  "name": "ProductOrder:grantAlteration",
  "function": "ProductOrder",
  "action": "grantAlteration",
  "specificationCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "reductionPercentage",
      "valueType": "number",
      "characteristicValueSpecification": [
        {
          "@type": "FloatCharacteristicValueSpecification",
          "valueType": "number",
          "value": 5.5
        }
      ]
    }
  ]
}
```

Example of a request for fully updating an entire permission specification.

Request

```
PATCH /permissionSpecification/ps-7e0a2d18
Content-Type: application/json
{
  "@type": "PermissionSpecification",
  "name": "ProductOrder:grantAlteration",
  "function": "ProductOrder",
```

```

    "action": "grantAlteration",
    "specificationCharacteristic": [
      {
        "@type": "CharacteristicSpecification",
        "name": "reductionPercentage",
        "valueType": "number",
        "characteristicValueSpecification": [
          {
            "@type": "FloatCharacteristicValueSpecification",
            "valueType": "number",
            "value": 5.5
          }
        ]
      }
    ]
  }
]
}

```

Response

```

200
{
  "@type": "PermissionSpecification",
  "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-7e0a2d18",
  "id": "ps-7e0a2d18",
  "name": "ProductOrder:grantAlteration",
  "function": "ProductOrder",
  "action": "grantAlteration",
  "specificationCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "reductionPercentage",
      "valueType": "number",
      "characteristicValueSpecification": [
        {
          "@type": "FloatCharacteristicValueSpecification",
          "valueType": "number",
          "value": 5.5
        }
      ]
    }
  ]
}

```

Example of a request for partial updating a permission specification, using JSON Patch Query.

Request

```

PATCH /permissionSpecification/ps-7e0a2d18
Content-Type: application/json-patch-query+json
[
  {
    "op": "replace",
    "path":
      "/specificationCharacteristic/characteristicValueSpecification/value?specificationCh

```

```
    "characteristic.name=reductionPercentage",
    "value": 5.5
  }
]
```

Response

```
200
{
  "@type": "PermissionSpecification",
  "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/permissionSpecification/ps-7e0a2d18",
  "id": "ps-7e0a2d18",
  "name": "ProductOrder:grantAlteration",
  "function": "ProductOrder",
  "action": "grantAlteration",
  "specificationCharacteristic": [
    {
      "@type": "CharacteristicSpecification",
      "name": "reductionPercentage",
      "valueType": "number",
      "characteristicValueSpecification": [
        {
          "@type": "FloatCharacteristicValueSpecification",
          "valueType": "number",
          "value": 5.5
        }
      ]
    }
  ]
}
```

Deletes a PermissionSpecification

DELETE /permissionSpecification/{id}

Description

This operation deletes a PermissionSpecification entity.

Usage samples

Here's an example of a request for delete a specific permission specification.

Request

```
DELETE /permissionSpecification/25632415
Content-Type: application/json
```

Response

204

Operations on CheckPermission

List or find CheckPermission objects

GET /checkPermission?fields=...

Description

This operation list CheckPermission entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve a list of succeeded check permission task resources.

Request

```
GET /checkPermission
Content-Type: application/json
```

Response

```
200
[
  {
    "id": "f3fc707ae9ce42fbbf273a46ccacdeca",
    "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/checkPermission/32434646753180"
    ,
    "@type": "CheckPermission",
    "state": "done",
    "permissionSpecification": {
      "@type": "PermissionSpecification",
      "function": "ProductOffering",
      "action": "create"
    },
    "user": {
      "role": "ProductManager",
      "@type": "RelatedSecurityPrincipal",
      "securityPrincipal": {
        "id": "Jonathan.Goldberg",
        "@type": "PartyRoleRef",
        "@referredType": "PartyRole"
      }
    }
  }
]
```

Retrieves a CheckPermission by ID

GET /checkPermission/{id}?fields=...

Description

This operation retrieves a CheckPermission entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Retrieve results of a check permission task by its identifier

Request

```
GET /checkPermission/x
Content-Type: application/json
```

Response

```
200
{
  "id": "f3fc707ae9ce42fbbf273a46ccacdeca",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/checkPermission/f3fc707ae9ce42fbbf273a46ccacdeca",
  "@type": "CheckPermission",
  "state": "done",
  "permissionSpecification": {
    "@type": "PermissionSpecification",
    "name": "ProductOffering::create",
    "function": "ProductOffering",
    "action": "create"
  },
  "user": {
    "role": "ProductManager",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "Jonathan.Goldberg",
      "@type": "PartyRoleRef",
      "@referredType": "PartyRole"
    }
  }
}
```

Initiate a task for CheckPermission

POST /checkPermission?fields=...

Description

This operation creates a CheckPermission entity.

Mandatory Attributes

Mandatory Attributes	Rule
permissionSpecification	
user	
@type	

Usage samples

Basic verification that the user is allowed to create any product offering.

Request

```
POST /checkPermission
Content-Type: application/json
{
  "@type": "CheckPermission",
  "permissionSpecification": {
    "@type": "PermissionSpecification",
    "name": "ProductOffering::create",
    "function": "ProductOffering",
    "action": "create"
  },
  "user": {
    "role": "ProductManager",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "Jonathan.Goldberg",
      "@type": "PartyRoleRef",
      "@referredType": "PartyRole"
    }
  }
}
```

Response

```
201
{
  "id": "f3fc707ae9ce42fbbf273a46ccacdeca",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/checkPermission/f3fc707ae9ce42fbbf273a46ccacdeca",
  "@type": "CheckPermission",
  "state": "InProgress",
  "permissionSpecification": {
    "@type": "PermissionSpecification",
    "name": "ProductOffering::create",
    "function": "ProductOffering",
    "action": "create"
  },
  "user": {
    "role": "ProductManager",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "Jonathan.Goldberg",
```

```

    "@type": "PartyRoleRef",
    "@referredType": "PartyRole"
  }
}
}

```

Verify that the user is allowed to give an ad-hoc price reduction of 10% in a product order.

Request

```

POST /checkPermission
Content-Type: application/json
{
  "@type": "CheckPermission",
  "permissionSpecification": {
    "@type": "PermissionSpecification",
    "name": "ProductOrder:grantAlteration",
    "function": "ProductOrder",
    "action": "grantAlteration"
  },
  "user": {
    "role": "SalesAgent",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "Jonathan.Goldberg",
      "@type": "PartyRoleRef",
      "@referredType": "PartyRole"
    }
  },
  "characteristic": [
    {
      "@type": "NumberCharacteristic",
      "name": "reductionPercentage",
      "valueType": "number",
      "value": 10
    }
  ],
  "entity": [
    {
      "id": "P0555333313",
      "@type": "EntityRef",
      "@referredType": "ProductOrder"
    }
  ]
}

```

Response

```

201
{
  "id": "bb58c1704de946a9bfe07f6d622287e5",
  "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/checkPermission/bb58c1704de946a9bfe07f6d622287e5",
  "@type": "CheckPermission",
  "state": "InProgress",
  "permissionSpecification": {

```

```

    "@type": "PermissionSpecification",
    "name": "ProductOrder:grantAlteration",
    "function": "ProductOrder",
    "action": "grantAlteration"
  },
  "user": {
    "role": "SalesAgent",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "Jonathan.Goldberg",
      "@type": "PartyRoleRef",
      "@referredType": "PartyRole"
    }
  },
  "characteristic": [
    {
      "@type": "NumberCharacteristic",
      "name": "reductionPercentage",
      "valueType": "number",
      "value": 10
    }
  ],
  "entity": [
    {
      "id": "P0555333313",
      "href":
"https://serverRoot/productOrderingManagement/v5/productOrder/P0555333313",
      "@type": "EntityRef",
      "@referredType": "ProductOrder"
    }
  ]
}

```

SteffiApf is a legal guardian for Hans, she needs permissions to read Hans' products, in this example product prod1

Request

```

POST /checkPermission
Content-Type: application/json
{
  "@type": "CheckPermission",
  "permissionSpecification": {
    "@type": "PermissionSpecification",
    "name": "Product:read",
    "function": "Product",
    "action": "read"
  },
  "user": {
    "role": "ProductUser",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "Jonathan.Goldberg",
      "@type": "PartyRoleRef",
      "@referredType": "PartyRole"
    }
  },
  "entity": [
    {
      "id": "g265-tf85",

```

```

    "@type": "EntityRef",
    "@referredType": "Product"
  }
]
}

```

Response

```

201
{
  "id": "687aa68dd7c24668a545ba00a071d2d5",
  "href":
  "https://serverRoot/rolesAndPermissionsManagement/v5/checkPermission/687aa68dd7c2466
  8a545ba00a071d2d5",
  "@type": "CheckPermission",
  "state": "InProgress",
  "permissionSpecification": {
    "@type": "PermissionSpecification",
    "name": "Product:read",
    "function": "Product",
    "action": "read"
  },
  "user": {
    "role": "ProductUser",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "Jonathan.Goldberg",
      "@type": "PartyRoleRef",
      "@referredType": "PartyRole"
    }
  },
  "entity": [
    {
      "id": "g265-tf85",
      "href": "https://serverRoot/productInventoryManagement/v5/product/g265-tf85",
      "@type": "EntityRef",
      "@referredType": "Product"
    }
  ]
}

```

User Jane Doe is a product marketer for mobile, she can create mobile offerings and prices, but not in other lines of business. No LoB explicit field in ProductOffering so using category to signify

Request

```

POST /checkPermission
Content-Type: application/json
{
  "@type": "CheckPermission",
  "permissionSpecification": {
    "@type": "PermissionSpecification",
    "name": "ProductOffering:read",
    "function": "ProductOffering",
    "action": "read"
  }
}

```

```

    },
    "user": {
      "role": "ProductMarketeer",
      "@type": "RelatedSecurityPrincipal",
      "securityPrincipal": {
        "id": "Jane.Doe",
        "@type": "PartyRoleRef",
        "@referredType": "PartyRole"
      }
    },
    "entity": [
      {
        "id": "7655",
        "@type": "EntityRef",
        "@referredType": "ProductOffering"
      }
    ]
  }
}

```

Response

```

201
{
  "id": "ba522cd932bc43ccb1f308c4bfff8ea3",
  "href":
    "https://serverRoot/rolesAndPermissionsManagement/v5/checkPermission/ba522cd932bc43c
    cb1f308c4bfff8ea3",
  "@type": "CheckPermission",
  "state": "InProgress",
  "permissionSpecification": {
    "@type": "PermissionSpecification",
    "name": "ProductOffering:read",
    "function": "ProductOffering",
    "action": "read"
  },
  "user": {
    "role": "ProductMarketeer",
    "@type": "RelatedSecurityPrincipal",
    "securityPrincipal": {
      "id": "Jane.Doe",
      "@type": "PartyRoleRef",
      "@referredType": "PartyRole"
    }
  },
  "entity": [
    {
      "id": "7655",
      "href": "https://serverRoot/productCatalogManagement/v5/productOffering/7655",
      "@type": "EntityRef",
      "@referredType": "ProductOffering"
    }
  ]
}

```

API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case **DELETE /api/hub/{id}** must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
  "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
  "id": "42",
  "callback": "http://in.listener.com",
  "query": ""
}
```

Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42
Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The **/client/listener** url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example **EVENT TYPE** should be replaced by one of the notification types supported by this API (see Notification resources Models section) and **EVENT BODY** refers to the data structure of the given notification type.

Request

```
POST /client/listener

Accept: application/json

{
  "event": {
    EVENT BODY
  },
  "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

Acknowledgements

Version History

Version Number	Date Modified	Modified by:	Description of changes
5.0.0	31-Jul-2021	Don Anton Dehipitiarachchi (Verizon), Jonathan Goldberg (Amdocs)	Rework for v5
5.0.0	28-Apr-2023	Jonathan Goldberg (Amdocs), Bruno Sousa Fernandes (NOS)	Added global party domain overview. Operation examples and use case enhancements.
5.0.1	29-Oct-2024	Bruno Sousa Fernandes (NOS)	Fixed examples according to new API tooling linter and two minor fixes in schema.
5.1.0	24-Oct-2025	Bruno Sousa Fernandes (NOS)	AP-4407 TMF672 Add permissions for Resource.

Release History

Release Number	Release Status	Date	Release led by:	Description
5.0.0	Preview	17-Jul-2024	Jonathan Goldberg (Amdocs)	Initial revision of the document.
5.0.1	Preview	14-Nov-2024	Bruno Sousa Fernandes (NOS)	OAS and User guide fixes including examples and two minor fixes in schema.

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